

Convex Optimization

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Sources:

- Boyd & Vandenberghe, *Convex Optimization*, 2004
- Courses EE236B, EE236C (UCLA), EE364A, EE364B (Stephen Boyd, Stanford Univ.)

Introduction

- mathematical optimization, modeling, complexity
- convex optimization
- recent history

Mathematical optimization

$$\begin{array}{ll}\text{minimize} & f_0(x_1, \dots, x_n) \\ \text{subject to} & f_1(x_1, \dots, x_n) \leq 0 \\ & \dots \\ & f_m(x_1, \dots, x_n) \leq 0\end{array}$$

- $x = (x_1, x_2, \dots, x_n)$ are decision variables
- $f_0(x_1, x_2, \dots, x_n)$ gives the cost of choosing x
- inequalities give constraints that x must satisfy

a mathematical model of a decision, design, or estimation problem

Limits of mathematical optimization

- how realistic is the model, and how certain are we about it?
- is the optimization problem tractable by existing numerical algorithms?

Optimization research

- **modeling**
generic techniques for formulating tractable optimization problems
- **algorithms**
expand class of problems that can be efficiently solved

Complexity of nonlinear optimization

- the general optimization problem is intractable
- even simple looking optimization problems can be very hard

Examples

- quadratic optimization problem with many constraints
- minimizing a multivariate polynomial

The famous exception: Linear programming

$$\begin{array}{ll} \text{minimize} & c^T x = \sum_{i=1}^n c_i x_i \\ \text{subject to} & a_i^T x \leq b_i, \quad i = 1, \dots, m \end{array}$$

- widely used since Dantzig introduced the simplex algorithm in 1948
- since 1950s, many applications in operations research, network optimization, finance, engineering, . . .
- extensive theory (optimality conditions, sensitivity, . . .)
- there exist very efficient algorithms for solving linear programs

Solving linear programs

- no closed-form expression for solution
- widely available and reliable software
- for some algorithms, can prove polynomial time
- problems with over 10^5 variables or constraints solved routinely

Convex optimization problem

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_1(x) \leq 0 \\ & \dots \\ & f_m(x) \leq 0\end{array}$$

- objective and constraint functions are convex: for $0 \leq \theta \leq 1$

$$f_i(\theta x + (1 - \theta)y) \leq \theta f_i(x) + (1 - \theta)f_i(y)$$

- includes least-squares problems and linear programs as special cases
- can be solved exactly, with similar complexity as LPs
- surprisingly many problems can be solved via convex optimization

History

- 1940s: linear programming

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & a_i^T x \leq b_i, \quad i = 1, \dots, m\end{array}$$

- 1950s: quadratic programming
- 1960s: geometric programming
- 1990s: semidefinite programming, second-order cone programming, quadratically constrained quadratic programming, robust optimization, sum-of-squares programming, . . .

New applications since 1990

- linear matrix inequality techniques in control
- circuit design via geometric programming
- support vector machine learning via quadratic programming
- semidefinite programming relaxations in combinatorial optimization
- ℓ_1 -norm optimization for sparse signal reconstruction
- applications in structural optimization, statistics, signal processing, communications, image processing, computer vision, quantum information theory, finance, . . .

Algorithms

Interior-point methods

- 1984 (Karmarkar): first practical polynomial-time algorithm
- 1984-1990: efficient implementations for large-scale LPs
- around 1990 (Nesterov & Nemirovski): polynomial-time interior-point methods for nonlinear convex programming
- since 1990: extensions and high-quality software packages

First-order algorithms

- similar to gradient descent, but with better convergence properties
- based on Nesterov's 'optimal' methods from 1980s
- extend to certain nondifferentiable or constrained problems

Outline

- basic theory
 - convex sets and functions
 - convex optimization problems
 - linear, quadratic, and geometric programming
- cone linear programming and applications
 - second-order cone programming
 - semidefinite programming
- some recent developments in algorithms (since 1990)
 - interior-point methods
 - fast gradient methods

Convex sets and functions

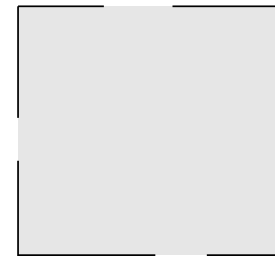
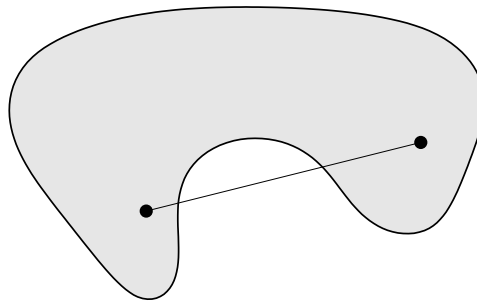
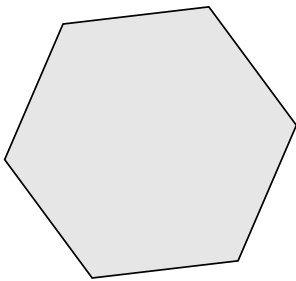
- definition
- basic examples and properties
- operations that preserve convexity

Convex set

contains line segment between any two points in the set

$$x_1, x_2 \in C, \quad 0 \leq \theta \leq 1 \quad \implies \quad \theta x_1 + (1 - \theta)x_2 \in C$$

Examples: one convex, two nonconvex sets



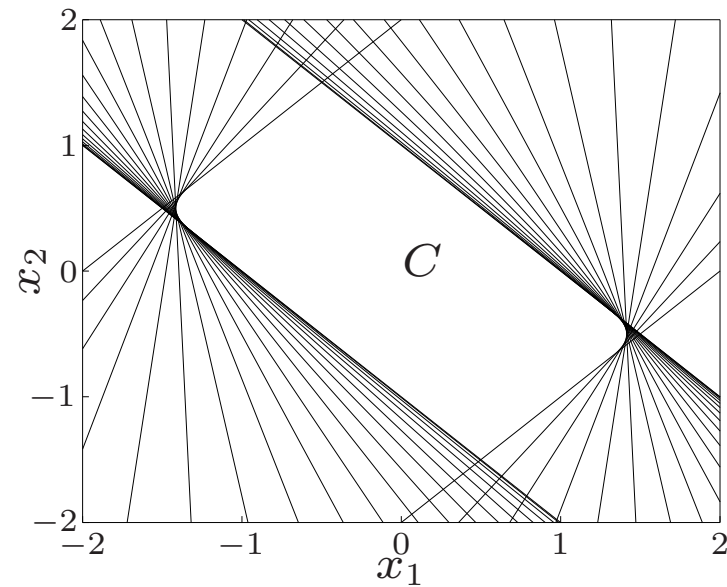
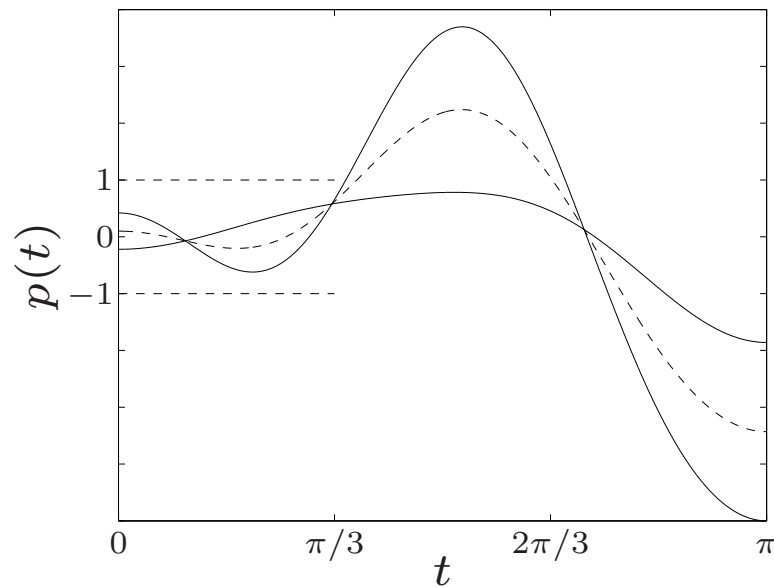
Examples and properties

- solution set of linear equations $Ax = b$ (affine set)
- solution set of linear inequalities $Ax \preceq b$ (polyhedron)
- norm balls $\{x \mid \|x\| \leq R\}$ and norm cones $\{(x, t) \mid \|x\| \leq t\}$
- set of positive semidefinite matrices $\mathbf{S}_+^n = \{X \in \mathbf{S}^n \mid X \succeq 0\}$
- image of a convex set under a linear transformation is convex
- inverse image of a convex set under a linear transformation is convex
- intersection of convex sets is convex

Example of intersection property

$$C = \{x \in \mathbf{R}^n \mid |p(t)| \leq 1 \text{ for } |t| \leq \pi/3\}$$

where $p(t) = x_1 \cos t + x_2 \cos 2t + \cdots + x_n \cos nt$



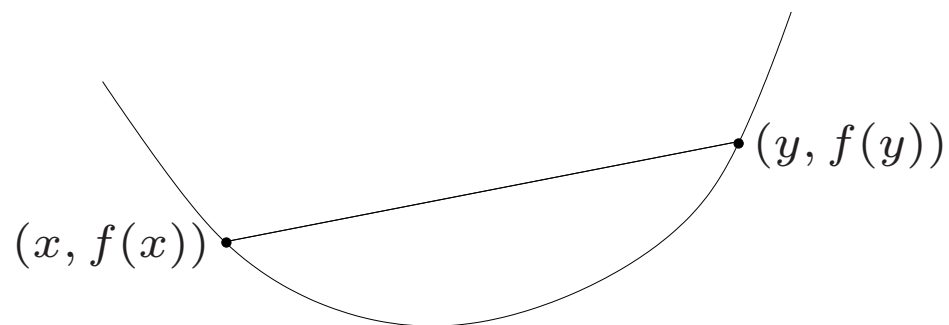
C is intersection of infinitely many halfspaces, hence convex

Convex function

domain $\text{dom } f$ is a convex set and

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$$

for all $x, y \in \text{dom } f$, $0 \leq \theta \leq 1$

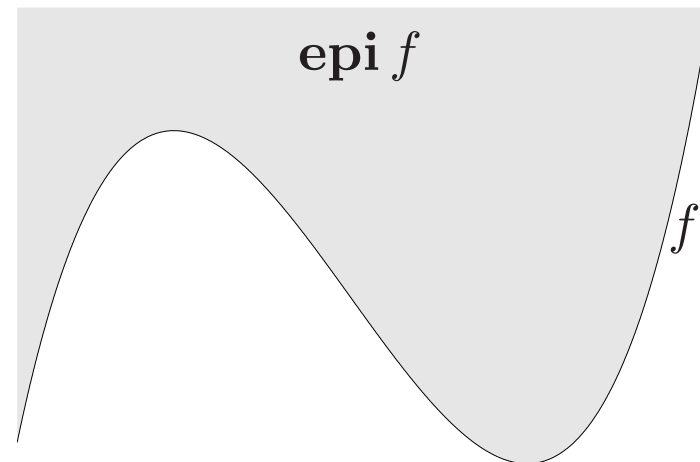


f is concave if $-f$ is convex

Epigraph and sublevel set

Epigraph: $\text{epi } f = \{(x, t) \mid x \in \text{dom } f, f(x) \leq t\}$

a function is convex if and only its epigraph is a convex set



Sublevel sets: $C_\alpha = \{x \in \text{dom } f \mid f(x) \leq \alpha\}$

the sublevel sets of a convex function are convex (converse is false)

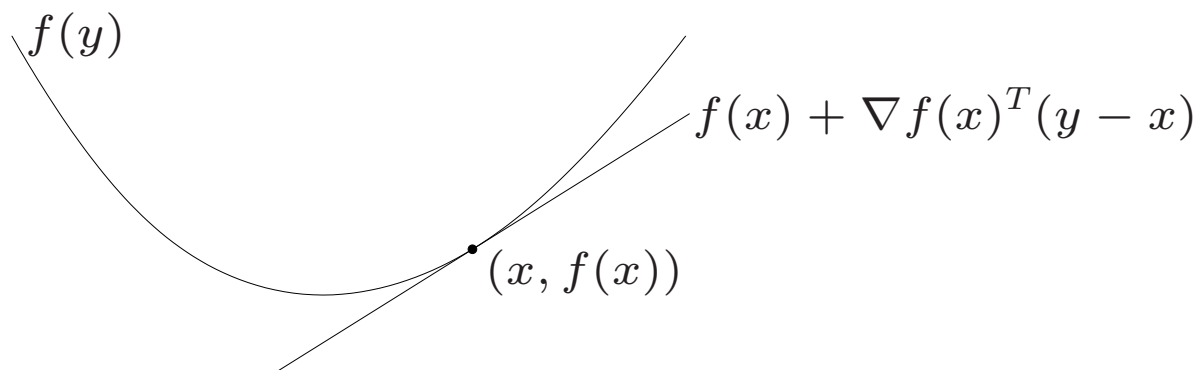
Examples

- $\exp x$, $-\log x$, $x \log x$ are convex
- x^α is convex for $x > 0$ and $\alpha \geq 1$ or $\alpha \leq 0$; $|x|^\alpha$ is convex for $\alpha \geq 1$
- quadratic-over-linear function $x^T x / t$ is convex in x, t for $t > 0$
- geometric mean $(x_1 x_2 \cdots x_n)^{1/n}$ is concave for $x \succeq 0$
- $\log \det X$ is concave on set of positive definite matrices
- $\log(e^{x_1} + \cdots e^{x_n})$ is convex
- linear and affine functions are convex and concave
- norms are convex

Differentiable convex functions

differentiable f is convex if and only if $\mathbf{dom} f$ is convex and

$$f(y) \geq f(x) + \nabla f(x)^T (y - x) \quad \text{for all } x, y \in \mathbf{dom} f$$



twice differentiable f is convex if and only if $\mathbf{dom} f$ is convex and

$$\nabla^2 f(x) \succeq 0 \quad \text{for all } x \in \mathbf{dom} f$$

Operations that preserve convexity

methods for establishing convexity of a function

1. verify definition
2. for twice differentiable functions, show $\nabla^2 f(x) \succeq 0$
3. show that f is obtained from simple convex functions by operations that preserve convexity
 - nonnegative weighted sum
 - composition with affine function
 - pointwise maximum and supremum
 - composition
 - minimization
 - perspective

Positive weighted sum & composition with affine function

Nonnegative multiple: αf is convex if f is convex, $\alpha \geq 0$

Sum: $f_1 + f_2$ convex if f_1, f_2 convex (extends to infinite sums, integrals)

Composition with affine function: $f(Ax + b)$ is convex if f is convex

Examples

- log barrier for linear inequalities

$$f(x) = - \sum_{i=1}^m \log(b_i - a_i^T x)$$

- (any) norm of affine function: $f(x) = \|Ax + b\|$

Pointwise maximum

$$f(x) = \max\{f_1(x), \dots, f_m(x)\}$$

is convex if $f_1, \dots, f_m$ are convex

Example: sum of r largest components of $x \in \mathbf{R}^n$

$$f(x) = x_{[1]} + x_{[2]} + \dots + x_{[r]}$$

is convex ($x_{[i]}$ is i th largest component of x)

proof:

$$f(x) = \max\{x_{i_1} + x_{i_2} + \dots + x_{i_r} \mid 1 \leq i_1 < i_2 < \dots < i_r \leq n\}$$

Pointwise supremum

$$g(x) = \sup_{y \in \mathcal{A}} f(x, y)$$

is convex if $f(x, y)$ is convex in x for each $y \in \mathcal{A}$

Example: maximum eigenvalue of symmetric matrix

$$\lambda_{\max}(X) = \sup_{\|y\|_2=1} y^T X y$$

Composition with scalar functions

composition of $g : \mathbf{R}^n \rightarrow \mathbf{R}$ and $h : \mathbf{R} \rightarrow \mathbf{R}$:

$$f(x) = h(g(x))$$

f is convex if

g convex, h convex and nondecreasing
 g concave, h convex and nonincreasing

(if we assign $h(x) = \infty$ for $x \in \mathbf{dom} h$)

Examples

- $\exp g(x)$ is convex if g is convex
- $1/g(x)$ is convex if g is concave and positive

Vector composition

composition of $g : \mathbf{R}^n \rightarrow \mathbf{R}^k$ and $h : \mathbf{R}^k \rightarrow \mathbf{R}$:

$$f(x) = h(g(x)) = h(g_1(x), g_2(x), \dots, g_k(x))$$

f is convex if

g_i convex, h convex and nondecreasing in each argument
 g_i concave, h convex and nonincreasing in each argument

(if we assign $h(x) = \infty$ for $x \in \mathbf{dom} h$)

Examples

- $\sum_{i=1}^m \log g_i(x)$ is concave if g_i are concave and positive
- $\log \sum_{i=1}^m \exp g_i(x)$ is convex if g_i are convex

Minimization

$$g(x) = \inf_{y \in C} f(x, y)$$

is convex if $f(x, y)$ is convex in x, y and C is a convex set

Examples

- distance to a convex set C : $g(x) = \inf_{y \in C} \|x - y\|$
- optimal value of linear program as function of righthand side

$$g(x) = \inf_{y: Ay \preceq x} c^T y$$

follows by taking

$$f(x, y) = c^T y, \quad \text{dom } f = \{(x, y) \mid Ay \preceq x\}$$

Perspective

the **perspective** of a function $f : \mathbf{R}^n \rightarrow \mathbf{R}$ is the function $g : \mathbf{R}^n \times \mathbf{R} \rightarrow \mathbf{R}$,

$$g(x, t) = tf(x/t)$$

g is convex if f is convex on $\text{dom } g = \{(x, t) \mid x/t \in \text{dom } f, t > 0\}$

Examples

- perspective of $f(x) = x^T x$ is quadratic-over-linear function

$$g(x, t) = \frac{x^T x}{t}$$

- perspective of negative logarithm $f(x) = -\log x$ is relative entropy

$$g(x, t) = t \log t - t \log x$$

Convex optimization problems

- standard form
- linear, quadratic, geometric programming
- modeling languages

Convex optimization problem

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & Ax = b\end{array}$$

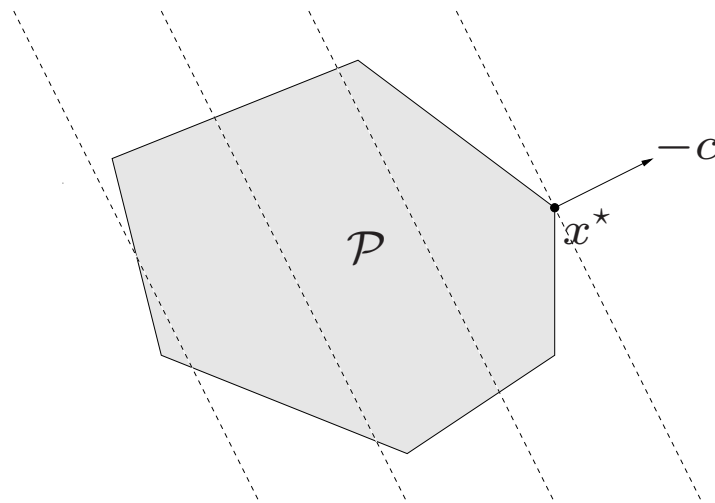
$f_0, f_1, \dots, f_m$ are convex functions

- feasible set is convex
- locally optimal points are globally optimal
- tractable, both in theory and practice

Linear program (LP)

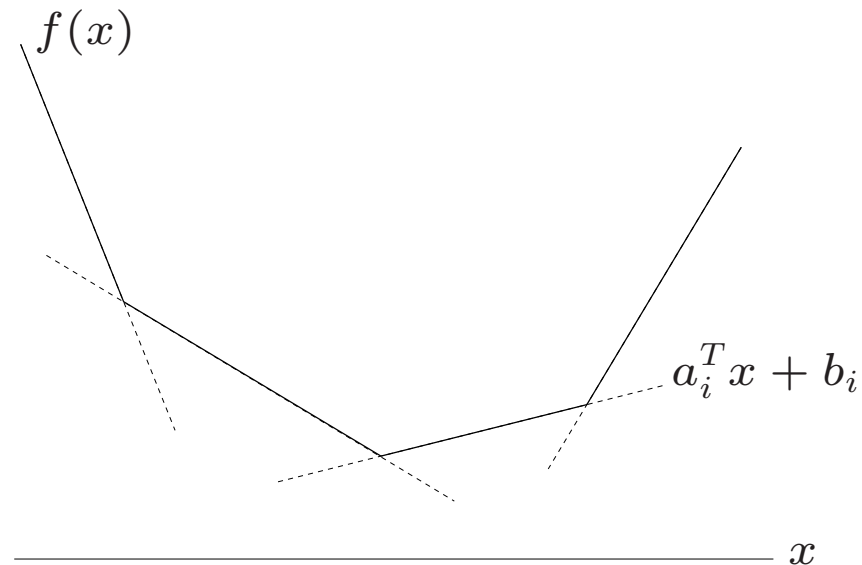
$$\begin{array}{ll}\text{minimize} & c^T x + d \\ \text{subject to} & Gx \preceq h \\ & Ax = b\end{array}$$

- inequality is componentwise vector inequality
- convex problem with affine objective and constraint functions
- feasible set is a polyhedron



Piecewise-linear minimization

$$\text{minimize } f(x) = \max_{i=1,\dots,m} (a_i^T x + b_i)$$



Equivalent linear program

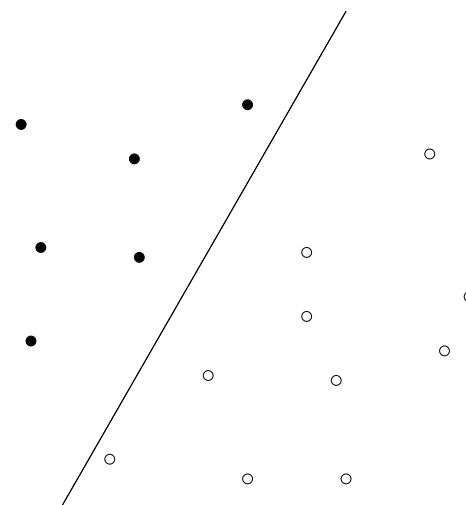
$$\begin{aligned} &\text{minimize} && t \\ &\text{subject to} && a_i^T x + b_i \leq t, \quad i = 1, \dots, m \end{aligned}$$

an LP with variables $x, t \in \mathbf{R}$

Linear discrimination

separate two sets of points $\{x_1, \dots, x_N\}$, $\{y_1, \dots, y_M\}$ by a hyperplane

$$\begin{aligned} a^T x_i + b &> 0, & i = 1, \dots, N \\ a^T y_i + b &< 0, & i = 1, \dots, M \end{aligned}$$

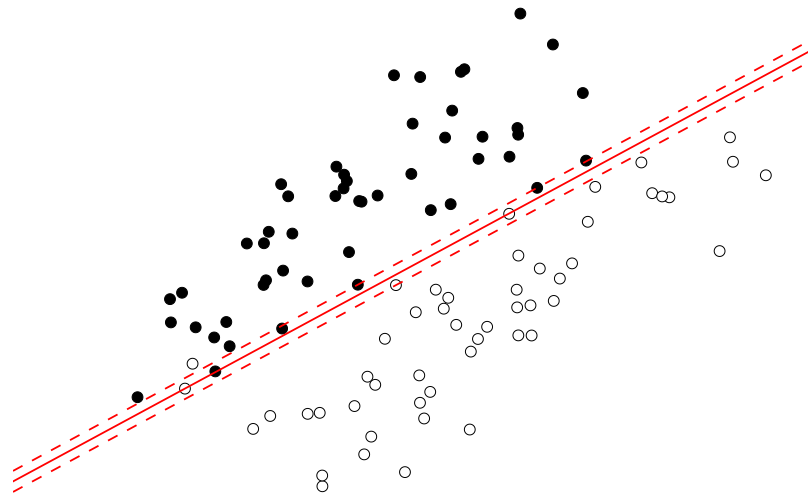


homogeneous in a , b , hence equivalent to the linear inequalities (in a , b)

$$a^T x_i + b \geq 1, \quad i = 1, \dots, N, \quad a^T y_i + b \leq -1, \quad i = 1, \dots, M$$

Approximate linear separation of non-separable sets

$$\text{minimize} \quad \sum_{i=1}^N \max\{0, 1 - a^T x_i - b\} + \sum_{i=1}^M \max\{0, 1 + a^T y_i + b\}$$



- a piecewise-linear minimization problem in a, b ; equivalent to an LP
- can be interpreted as a heuristic for minimizing #misclassified points

ℓ_1 -Norm and ℓ_∞ -norm minimization

ℓ_1 -Norm approximation and equivalent LP ($\|y\|_1 = \sum_k |y_k|$)

$$\text{minimize} \quad \|Ax - b\|_1$$

$$\begin{aligned} &\text{minimize} \quad \sum_{i=1}^n y_i \\ &\text{subject to} \quad -y \preceq Ax - b \preceq y \end{aligned}$$

ℓ_∞ -Norm approximation ($\|y\|_\infty = \max_k |y_k|$)

$$\text{minimize} \quad \|Ax - b\|_\infty$$

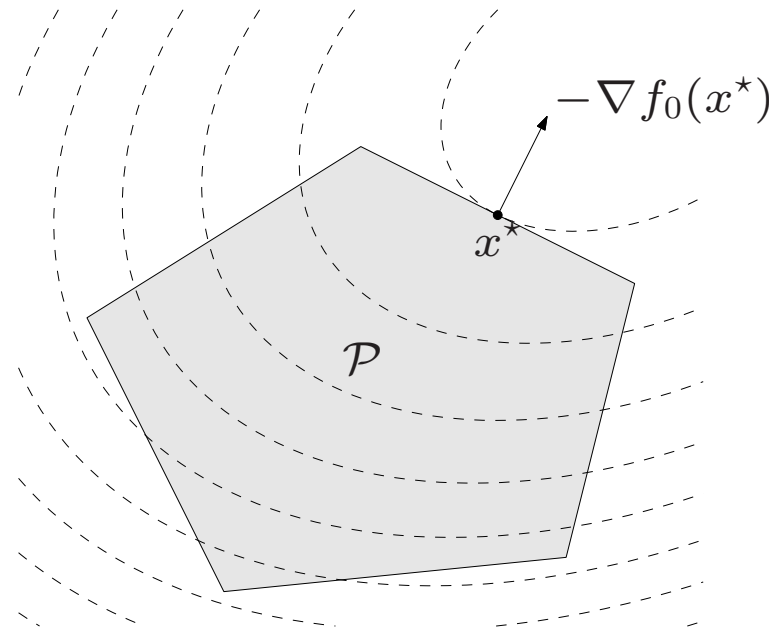
$$\begin{aligned} &\text{minimize} \quad y \\ &\text{subject to} \quad -y\mathbf{1} \preceq Ax - b \preceq y\mathbf{1} \end{aligned}$$

($\mathbf{1}$ is vector of ones)

Quadratic program (QP)

$$\begin{array}{ll}\text{minimize} & (1/2)x^T P x + q^T x + r \\ \text{subject to} & Gx \preceq h \\ & Ax = b\end{array}$$

- $P \in \mathbf{S}_{+}^n$, so objective is convex quadratic
- minimize a convex quadratic function over a polyhedron



Linear program with random cost

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & Gx \preceq h\end{array}$$

- c is random vector with mean $\bar{c}$ and covariance Σ
- hence, $c^T x$ is random variable with mean $\bar{c}^T x$ and variance $x^T \Sigma x$

Expected cost-variance trade-off

$$\begin{array}{ll}\text{minimize} & \mathbf{E} c^T x + \gamma \mathbf{var}(c^T x) = \bar{c}^T x + \gamma x^T \Sigma x \\ \text{subject to} & Gx \preceq h\end{array}$$

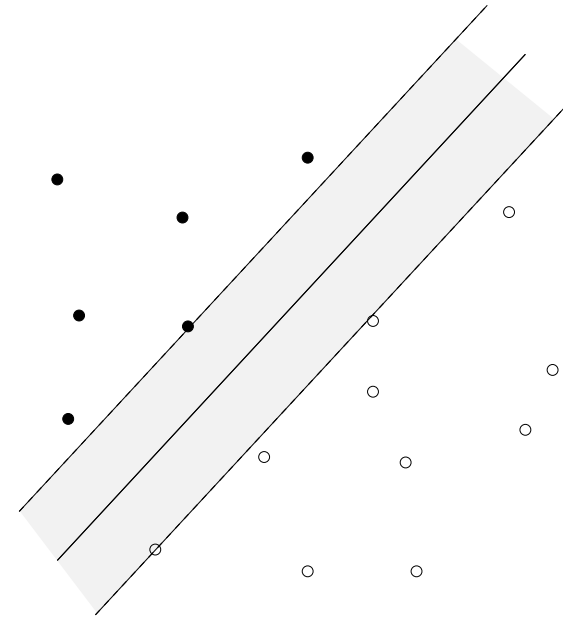
$\gamma > 0$ is risk aversion parameter

Robust linear discrimination

$$\mathcal{H}_1 = \{z \mid a^T z + b = 1\}$$

$$\mathcal{H}_2 = \{z \mid a^T z + b = -1\}$$

distance between hyperplanes is $2/\|a\|_2$



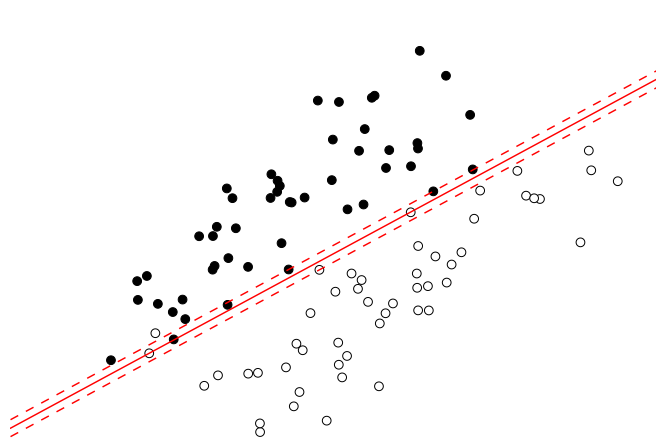
to separate two sets of points by maximum margin,

$$\begin{aligned} & \text{minimize} && \|a\|_2^2 = a^T a \\ & \text{subject to} && a^T x_i + b \geq 1, \quad i = 1, \dots, N \\ & && a^T y_i + b \leq -1, \quad i = 1, \dots, M \end{aligned}$$

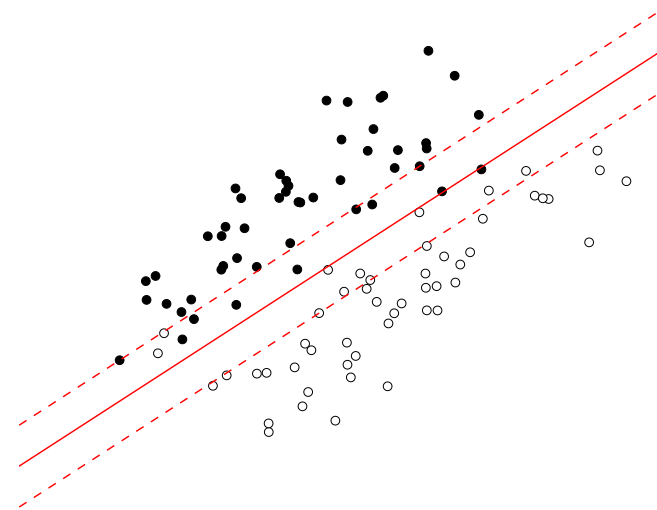
a quadratic program in a, b

Support vector classifier

$$\min. \quad \gamma \|a\|_2^2 + \sum_{i=1}^N \max\{0, 1 - a^T x_i - b\} + \sum_{i=1}^M \max\{0, 1 + a^T y_i + b\}$$



$\gamma = 0$

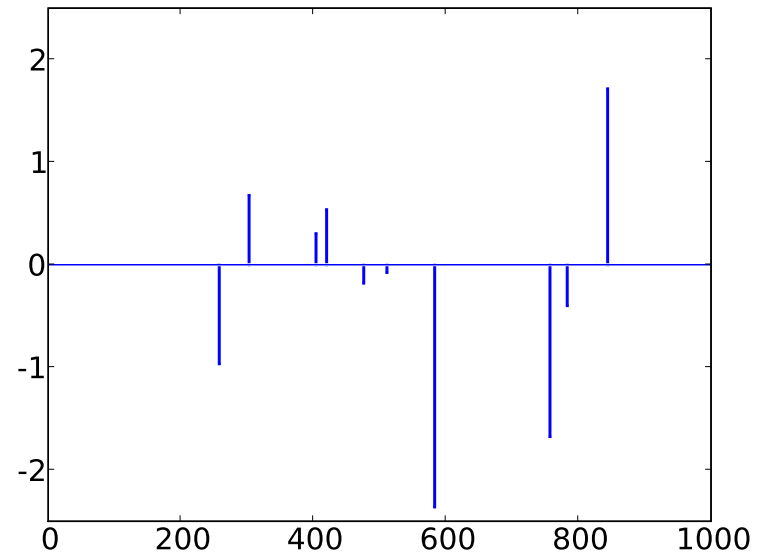


$\gamma = 10$

equivalent to a QP

Sparse signal reconstruction

- signal $\hat{x}$ of length 1000
- ten nonzero components



reconstruct signal from $m = 100$ random noisy measurements

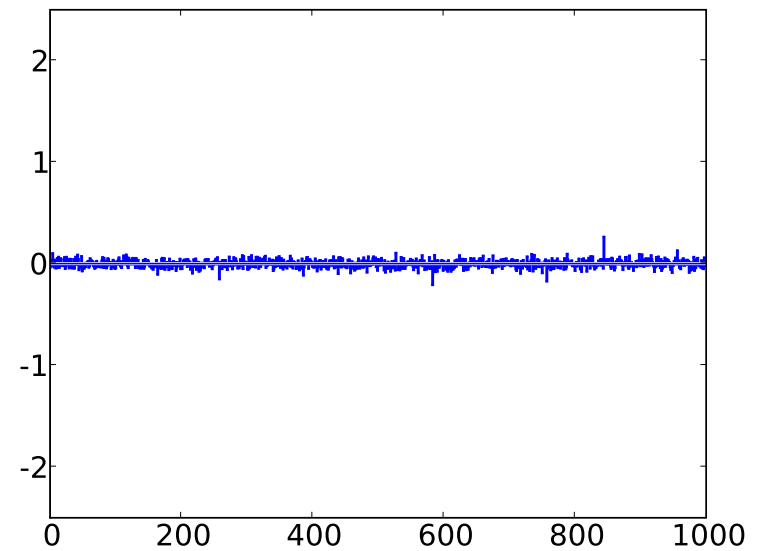
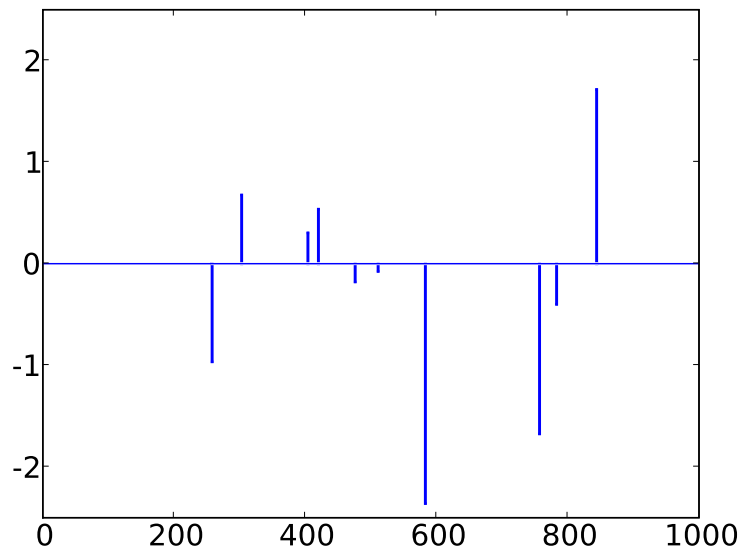
$$b = A\hat{x} + v$$

($A_{ij} \sim \mathcal{N}(0, 1)$ i.i.d. and $v \sim \mathcal{N}(0, \sigma^2 I)$ with $\sigma = 0.01$)

ℓ_2 -Norm regularization

$$\text{minimize} \quad \|Ax - b\|_2^2 + \gamma \|x\|_2^2$$

a least-squares problem

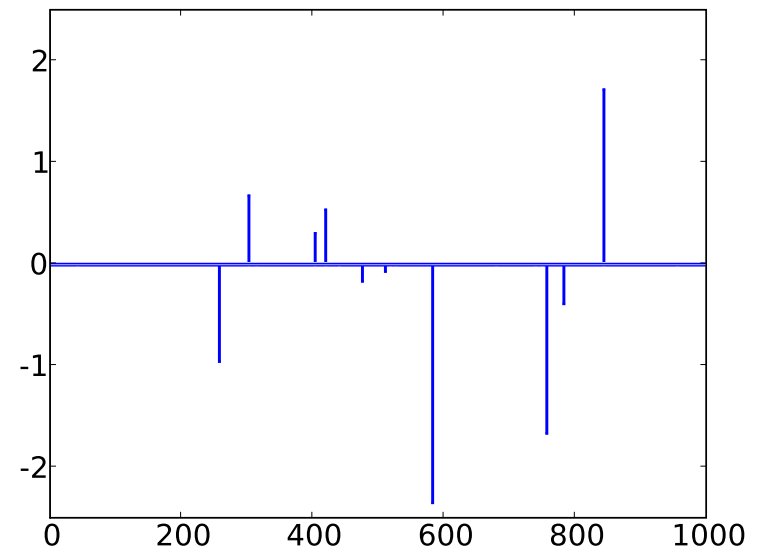
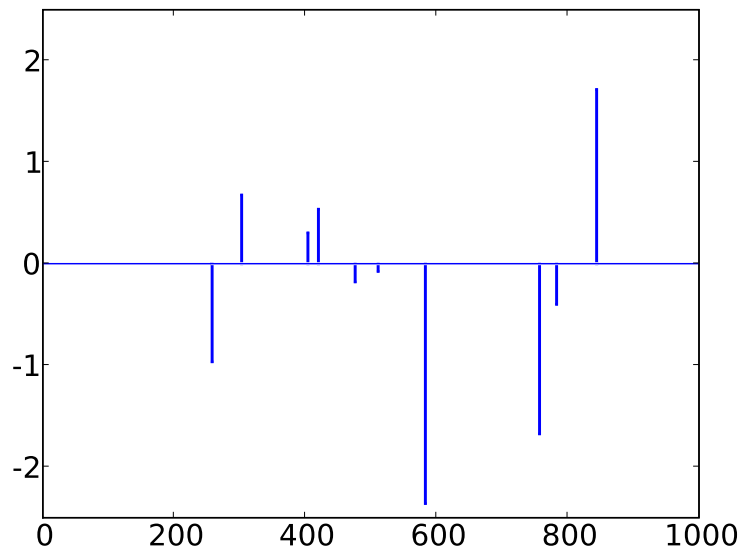


left: exact signal $\hat{x}$; right: 2-norm reconstruction

ℓ_1 -Norm regularization

$$\text{minimize} \quad \|Ax - b\|_2^2 + \gamma \|x\|_1$$

equivalent to a QP



left: exact signal $\hat{x}$; right: 1-norm reconstruction

Geometric programming

Posynomial function:

$$f(x) = \sum_{k=1}^K c_k x_1^{a_{1k}} x_2^{a_{2k}} \cdots x_n^{a_{nk}}, \quad \text{dom } f = \mathbf{R}_{++}^n$$

with $c_k > 0$

Geometric program (GP)

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 1, \quad i = 1, \dots, m \end{array}$$

with f_i posynomial

Geometric program in convex form

change variables to

$$y_i = \log x_i,$$

and take logarithm of cost, constraints

Geometric program in convex form:

$$\begin{aligned} \text{minimize} \quad & \log \left(\sum_{k=1}^K \exp(a_{0k}^T y + b_{0k}) \right) \\ \text{subject to} \quad & \log \left(\sum_{k=1}^K \exp(a_{ik}^T y + b_{ik}) \right) \leq 0, \quad i = 1, \dots, m \end{aligned}$$

$$b_{ik} = \log c_{ik}$$

Modeling software

Modeling packages for convex optimization

- CVX, Yalmip (Matlab)
- CVXMOD (Python)

assist in formulating convex problems by automating two tasks:

- verifying convexity from convex calculus rules
- transforming problem in input format required by standard solvers

Related packages

general purpose optimization modeling: AMPL, GAMS

CVX example

$$\begin{array}{ll}\text{minimize} & \|Ax - b\|_1 \\ \text{subject to} & -0.5 \leq x_k \leq 0.3, \quad k = 1, \dots, n\end{array}$$

Matlab code

```
A = randn(5, 3);  b = randn(5, 1);
cvx_begin
    variable x(3);
    minimize(norm(A*x - b, 1))
    subject to
        -0.5 <= x;
        x <= 0.3;
cvx_end
```

- between `cvx_begin` and `cvx_end`, `x` is a CVX variable
- after execution, `x` is Matlab variable with optimal solution

Cone programming

- generalized inequalities
- second-order cone programming
- semidefinite programming

Cone linear program

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & Gx \preceq_K h \\ & Ax = b\end{array}$$

- $y \preceq_K z$ means $z - y \in K$, where K is a proper convex cone
- extends linear programming ($K = \mathbf{R}_+^m$) to nonpolyhedral cones
- popular as standard format for nonlinear convex optimization
- theory and algorithms very similar to linear programming

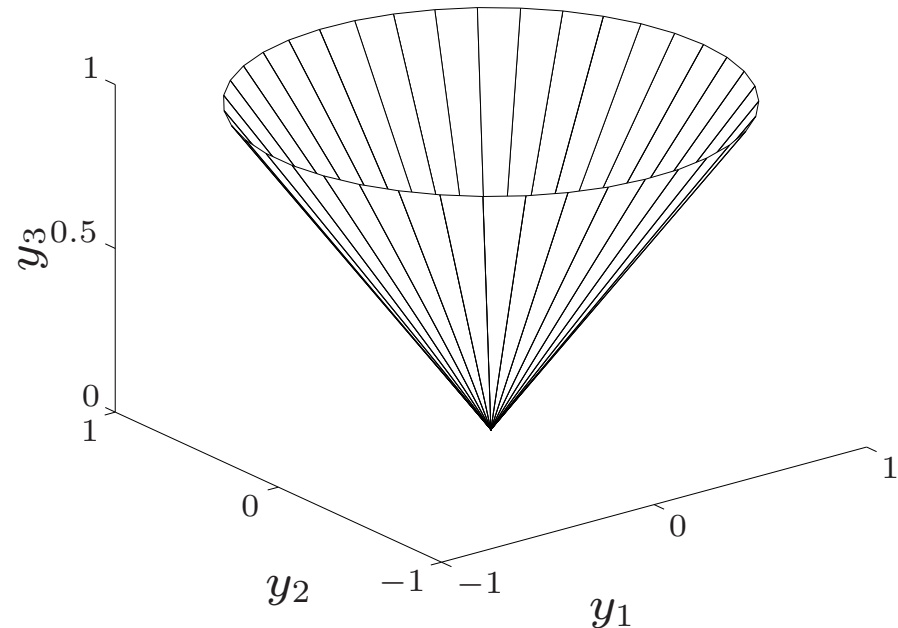
Second-order cone program (SOCP)

$$\begin{array}{ll}\text{minimize} & f^T x \\ \text{subject to} & \|A_i x + b_i\|_2 \leq c_i^T x + d_i, \quad i = 1, \dots, m\end{array}$$

- $\|\cdot\|_2$ is Euclidean norm $\|y\|_2 = \sqrt{y_1^2 + \dots + y_n^2}$
- constraints are nonlinear, nondifferentiable, convex

constraints are inequalities
w.r.t. second-order cone:

$$\left\{ y \mid \sqrt{y_1^2 + \dots + y_{p-1}^2} \leq y_p \right\}$$



Examples of SOC-representable constraints

Convex quadratic constraint ($A = LL^T$ positive definite)

$$x^T A x + 2b^T x + c \leq 0 \quad \Longleftrightarrow \quad \|L^T x + L^{-1}b\|_2 \leq (b^T A^{-1}b - c)^{1/2}$$

also extends to positive semidefinite singular A

Hyperbolic constraint

$$x^T x \leq yz, \quad y, z \geq 0 \quad \Longleftrightarrow \quad \left\| \begin{bmatrix} 2x \\ y - z \end{bmatrix} \right\|_2 \leq y + z, \quad y, z \geq 0$$

Examples of SOC-representable constraints

Positive powers

$$x^{1.5} \leq t, \quad x \geq 0 \quad \Longleftrightarrow \quad \exists z : \quad x^2 \leq tz, \quad z^2 \leq x, \quad x, z \geq 0$$

- two hyperbolic constraints can be converted to SOC constraints
- extends to powers x^p for rational $p \geq 1$

Negative powers

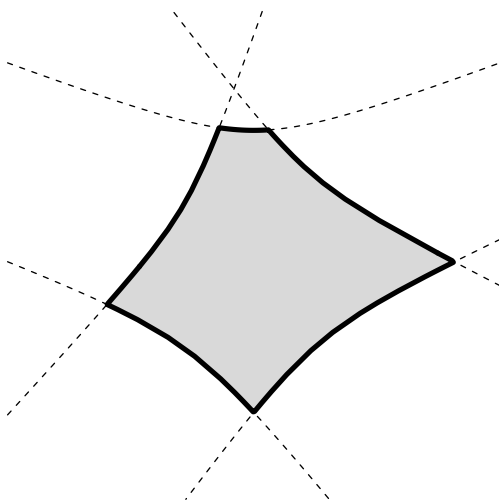
$$x^{-3} \leq t, \quad x > 0 \quad \Longleftrightarrow \quad \exists z : \quad 1 \leq tz, \quad z^2 \leq tx, \quad x, z \geq 0$$

- two hyperbolic constraints can be converted to SOC constraints
- extends to powers x^p for rational $p < 0$

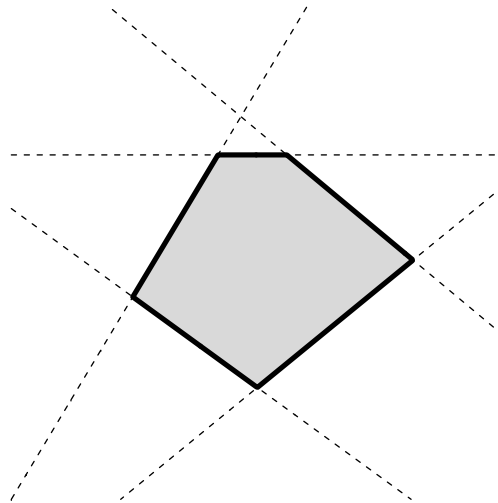
Robust linear program (stochastic)

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & \mathbf{prob}(a_i^T x \leq b_i) \geq \eta, \quad i = 1, \dots, m\end{array}$$

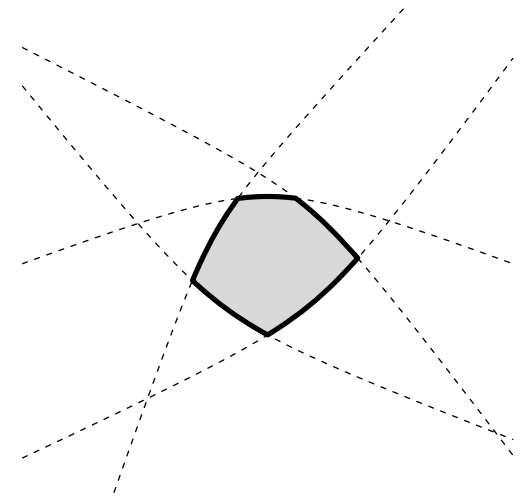
- a_i random and normally distributed with mean $\bar{a}_i$, covariance Σ_i
- we require that x satisfies each constraint with probability exceeding η



$\eta = 10\%$



$\eta = 50\%$



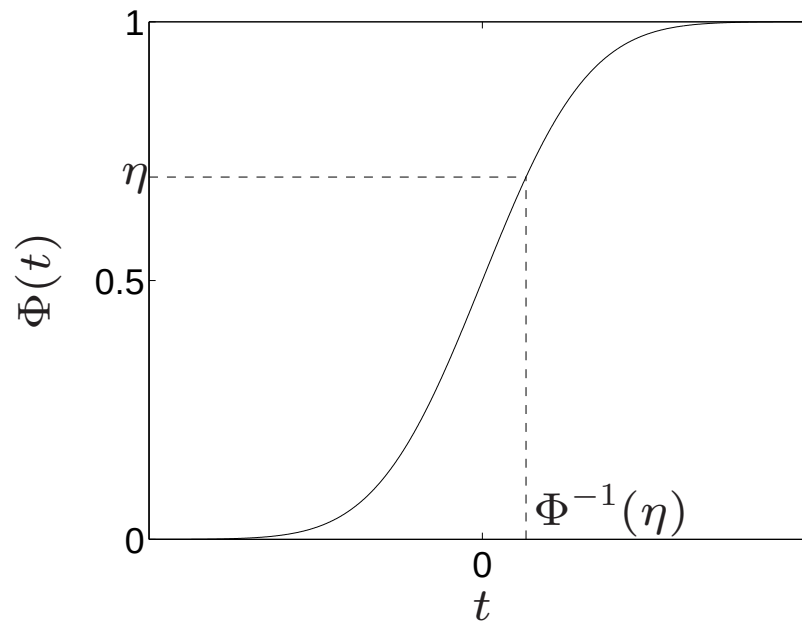
$\eta = 90\%$

SOCP formulation

the 'chance constraint' $\text{prob}(a_i^T x \leq b_i) \geq \eta$ is equivalent to the constraint

$$\bar{a}_i^T x + \Phi^{-1}(\eta) \|\Sigma_i^{1/2} x\|_2 \leq b_i$$

Φ is the (unit) normal cumulative density function



robust LP is a second-order cone program for $\eta \geq 0.5$

Robust linear program (deterministic)

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & a_i^T x \leq b_i \text{ for all } a_i \in \mathcal{E}_i, \quad i = 1, \dots, m\end{array}$$

- a_i uncertain but bounded by ellipsoid $\mathcal{E}_i = \{\bar{a}_i + P_i u \mid \|u\|_2 \leq 1\}$
- we require that x satisfies each constraint for all possible a_i

SOCP formulation

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & \bar{a}_i^T x + \|P_i^T x\|_2 \leq b_i, \quad i = 1, \dots, m\end{array}$$

follows from

$$\sup_{\|u\|_2 \leq 1} (\bar{a}_i + P_i u)^T x = \bar{a}_i^T x + \|P_i^T x\|_2$$

Semidefinite program (SDP)

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & x_1 A_1 + x_2 A_2 + \cdots + x_n A_n \preceq B\end{array}$$

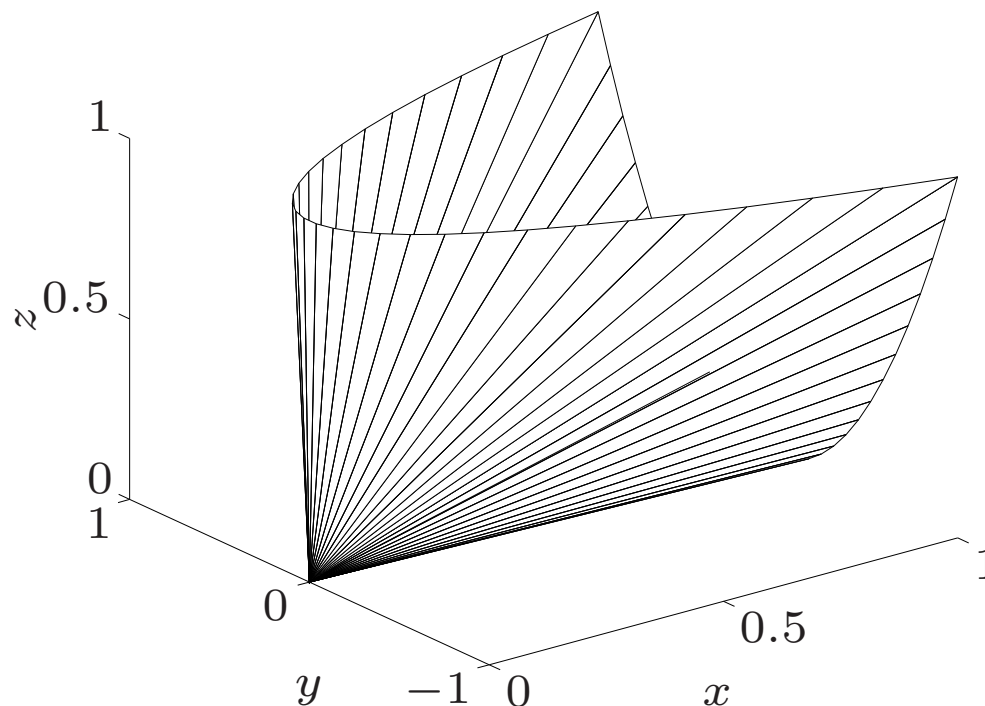
- $A_1, A_2, \dots, A_n, B$ are symmetric matrices
- inequality $X \preceq Y$ means $Y - X$ is *positive semidefinite*, i.e.,

$$z^T (Y - X) z = \sum_{i,j} (Y_{ij} - X_{ij}) z_i z_j \geq 0 \text{ for all } z$$

- includes many nonlinear constraints as special cases

Geometry

$$\begin{bmatrix} x & y \\ y & z \end{bmatrix} \succeq 0$$



- a nonpolyhedral convex cone
- feasible set of a semidefinite program is the intersection of the positive semidefinite cone in high dimension with planes

Examples

$$A(x) = A_0 + x_1 A_1 + \cdots + x_m A_m \quad (A_i \in \mathbf{S}^n)$$

Eigenvalue minimization (and equivalent SDP)

$$\text{minimize} \quad \lambda_{\max}(A(x))$$

$$\begin{array}{ll} \text{minimize} & t \\ \text{subject to} & A(x) \preceq tI \end{array}$$

Matrix-fractional function

$$\begin{array}{ll} \text{minimize} & b^T A(x)^{-1} b \\ \text{subject to} & A(x) \succeq 0 \end{array}$$

$$\begin{array}{ll} \text{minimize} & t \\ \text{subject to} & \begin{bmatrix} A(x) & b \\ b^T & t \end{bmatrix} \succeq 0 \end{array}$$

Matrix norm minimization

$$A(x) = A_0 + x_1 A_1 + x_2 A_2 + \cdots + x_n A_n \quad (A_i \in \mathbf{R}^{p \times q})$$

Matrix norm approximation ($\|X\|_2 = \max_k \sigma_k(X)$)

$$\text{minimize} \quad \|A(x)\|_2$$

$$\begin{array}{ll} \text{minimize} & t \\ \text{subject to} & \begin{bmatrix} tI & A(x)^T \\ A(x) & tI \end{bmatrix} \succeq 0 \end{array}$$

Nuclear norm approximation ($\|X\|_* = \sum_k \sigma_k(X)$)

$$\text{minimize} \quad \|A(x)\|_*$$

$$\begin{array}{ll} \text{minimize} & (\text{tr } U + \text{tr } V)/2 \\ \text{subject to} & \begin{bmatrix} U & A(x)^T \\ A(x) & V \end{bmatrix} \succeq 0 \end{array}$$

Semidefinite relaxations & randomization

semidefinite programming is increasingly used

- to find good bounds for hard (i.e., nonconvex) problems, via **relaxation**
- as a heuristic for good suboptimal points, often via **randomization**

Example: Boolean least-squares

$$\begin{array}{ll}\text{minimize} & \|Ax - b\|_2^2 \\ \text{subject to} & x_i^2 = 1, \quad i = 1, \dots, n\end{array}$$

- basic problem in digital communications
- could check all 2^n possible values of $x \in \{-1, 1\}^n \dots$
- an NP-hard problem, and very hard in practice

Semidefinite lifting

with $P = A^T A$, $q = -A^T b$, $r = b^T b$

$$\|Ax - b\|_2^2 = \sum_{i,j=1}^n P_{ij} x_i x_j + 2 \sum_{i=1}^n q_i x_i + r$$

after introducing new variables $X_{ij} = x_i x_j$

$$\begin{aligned} &\text{minimize} && \sum_{i,j=1}^n P_{ij} X_{ij} + 2 \sum_{i=1}^n q_i x_i + r \\ &\text{subject to} && X_{ii} = 1, \quad i = 1, \dots, n \\ &&& X_{ij} = x_i x_j, \quad i, j = 1, \dots, n \end{aligned}$$

- cost function and first constraints are linear
- last constraint in matrix form is $X = xx^T$, nonlinear and nonconvex,
... still a very hard problem

Semidefinite relaxation

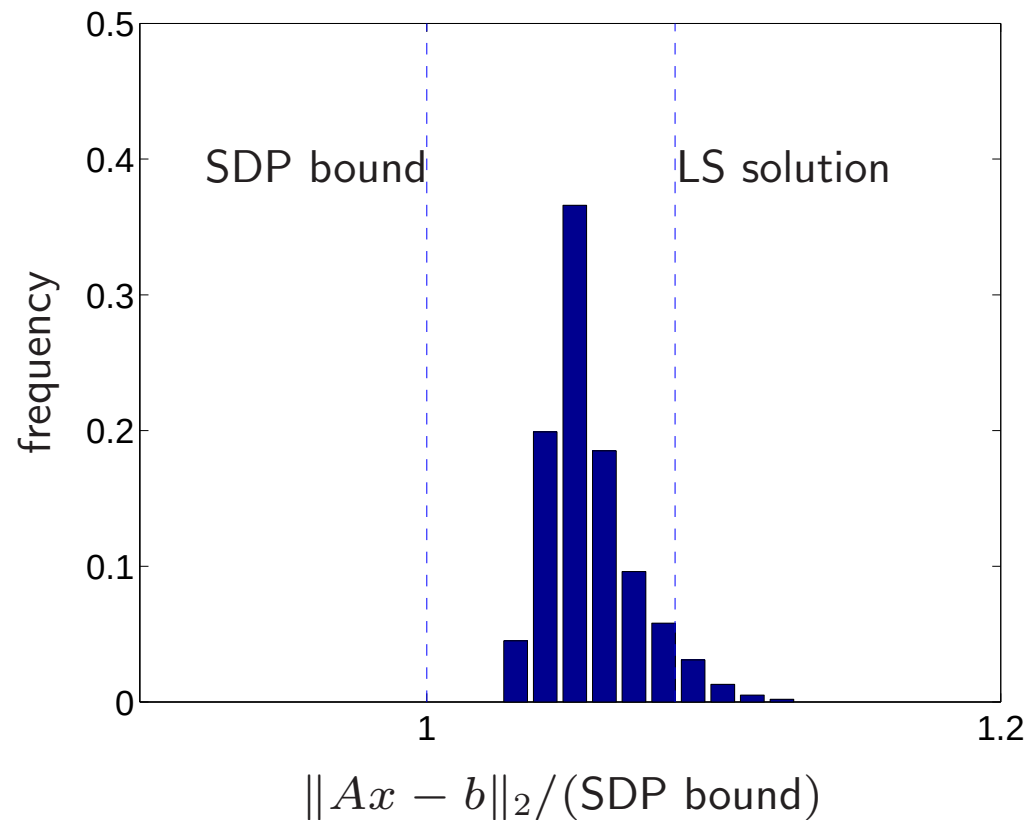
replace $X = xx^T$ with weaker constraint $X \succeq xx^T$, to obtain **relaxation**

$$\begin{aligned} & \text{minimize} && \sum_{i,j=1}^n P_{ij} X_{ij} + 2 \sum_{i=1}^n q_i x_i + r \\ & \text{subject to} && X_{ii} = 1, \quad i = 1, \dots, n \\ & && X \succeq xx^T \end{aligned}$$

- convex; can be solved as an semidefinite program
- optimal value gives lower bound for BLS
- if $X = xx^T$ at the optimum, we have solved the exact problem
- otherwise, can use *randomized rounding*

generate z from $\mathcal{N}(x, X - xx^T)$ and take $x = \mathbf{sign}(z)$

Example



- feasible set has $2^{100} \approx 10^{30}$ points
- histogram of 1000 randomized solutions from SDP relaxation

Nonnegative polynomial on $\mathbf{R}$

$$f(t) = x_0 + x_1 t + \cdots + x_{2m} t^{2m} \geq 0 \quad \text{for all } t \in \mathbf{R}$$

- a convex constraint on x
- holds if and only if f is a sum of squares of (two) polynomials:

$$\begin{aligned} f(t) &= \sum_k (y_{k0} + y_{k1}t + \cdots + y_{km}t^m)^2 \\ &= \begin{bmatrix} 1 \\ \vdots \\ t^m \end{bmatrix}^T \sum_k y_k y_k^T \begin{bmatrix} 1 \\ \vdots \\ t^m \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ \vdots \\ t^m \end{bmatrix}^T Y \begin{bmatrix} 1 \\ \vdots \\ t^m \end{bmatrix} \end{aligned}$$

where $Y = \sum_k y_k y_k^T \succeq 0$

SDP formulation

$f(t) \geq 0$ if and only if for some $Y \succeq 0$,

$$f(t) = \begin{bmatrix} 1 \\ t \\ \vdots \\ t^{2m} \end{bmatrix}^T \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{2m} \end{bmatrix} = \begin{bmatrix} 1 \\ t \\ \vdots \\ t^m \end{bmatrix}^T Y \begin{bmatrix} 1 \\ t \\ \vdots \\ t^m \end{bmatrix}$$

this is an SDP constraint: there exists $Y \succeq 0$ such that

$$\begin{aligned} x_0 &= Y_{11} \\ x_1 &= Y_{12} + Y_{21} \\ x_2 &= Y_{13} + Y_{22} + Y_{32} \\ &\vdots \\ x_{2m} &= Y_{m+1,m+1} \end{aligned}$$

General sum-of-squares constraints

$f(t) = x^T p(t)$ is a sum of squares if

$$x^T p(t) = \sum_{k=1}^s (y_k^T q(t))^2 = q(t)^T \left(\sum_{k=1}^s y_k y_k^T \right) q(t)$$

- p, q : basis functions (of polynomials, trigonometric polynomials, . . .)
- independent variable t can be one- or multidimensional
- a *sufficient* condition for nonnegativity of $x^T p(t)$, useful in nonconvex polynomial optimization in several variables
- in some nontrivial cases (*e.g.*, polynomial on $\mathbf{R}$), *necessary and sufficient*

Equivalent SDP constraint (on the variables x, X)

$$x^T p(t) = q(t)^T X q(t), \quad X \succeq 0$$

Example: Cosine polynomials

$$f(\omega) = x_0 + x_1 \cos \omega + \cdots + x_{2n} \cos 2n\omega \geq 0$$

Sum of squares theorem: $f(\omega) \geq 0$ for $\alpha \leq \omega \leq \beta$ if and only if

$$f(\omega) = g_1(\omega)^2 + s(\omega)g_2(\omega)^2$$

- g_1, g_2 : cosine polynomials of degree n and $n - 1$
- $s(\omega) = (\cos \omega - \cos \beta)(\cos \alpha - \cos \omega)$ is a given weight function

Equivalent SDP formulation: $f(\omega) \geq 0$ for $\alpha \leq \omega \leq \beta$ if and only if

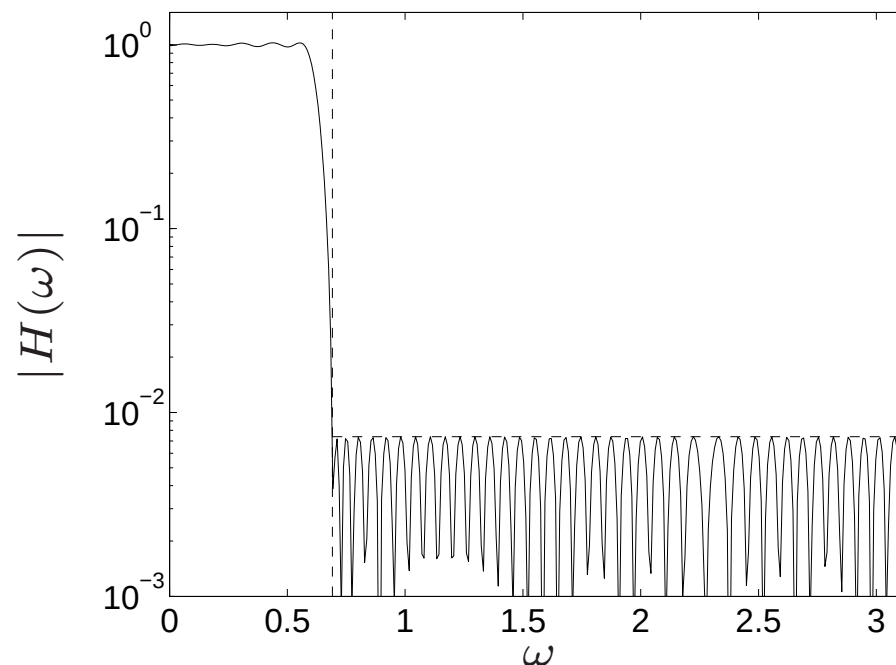
$$x^T p(\omega) = q_1(\omega)^T X_1 q_1(\omega) + s(\omega) q_2(\omega)^T X_2 q_2(\omega), \quad X_1 \succeq 0, \quad X_2 \succeq 0$$

p, q_1, q_2 : basis vectors $(1, \cos \omega, \cos(2\omega), \dots)$ up to order $2n, n, n - 1$

Example: Linear-phase Nyquist filter

$$\text{minimize} \quad \sup_{\omega \geq \omega_s} |h_0 + h_1 \cos \omega + \cdots + h_{2n} \cos 2n\omega|$$

with $h_0 = 1/M$, $h_{kM} = 0$ for positive integer k



(Example with $n = 25$, $M = 5$, $\omega_s = 0.69$)

SDP formulation

$$\begin{array}{ll}\text{minimize} & t \\ \text{subject to} & -t \leq H(\omega) \leq t, \quad \omega_s \leq \omega \leq \pi\end{array}$$

where $H(\omega) = h_0 + h_1 \cos \omega + \cdots + h_{2n} \cos 2n\omega$

Equivalent SDP

$$\begin{array}{ll}\text{minimize} & t \\ \text{subject to} & t - H(\omega) = q_1(\omega)^T X_1 q_1(\omega) + s(\omega) q_2(\omega)^T X_2 q_2(\omega) \\ & t + H(\omega) = q_1(\omega)^T X_3 q_1(\omega) + s(\omega) q_2(\omega)^T X_4 q_2(\omega) \\ & X_1 \succeq 0, \quad X_2 \succeq 0, \quad X_3 \succeq 0, \quad X_4 \succeq 0\end{array}$$

Variables t , h_i ($i \neq kM$), 4 matrices X_i of size roughly n

Chebyshev inequalities

Classical (two-sided) Chebyshev inequality

$$\text{prob}(|X| < 1) \geq 1 - \sigma^2$$

- holds for all random X with $\mathbf{E} X = 0$, $\mathbf{E} X^2 = \sigma^2$
- there exists a distribution that achieves the bound

Generalized Chebyshev inequalities

give lower bound on $\text{prob}(X \in C)$, given moments of X

Chebyshev inequality for quadratic constraints

- C is defined by quadratic inequalities

$$C = \{x \in \mathbf{R}^n \mid x^T A_i x + 2b_i^T x + c_i \leq 0, \ i = 1, \dots, m\}$$

- X is random vector with $\mathbf{E} X = a$, $\mathbf{E} X X^T = S$

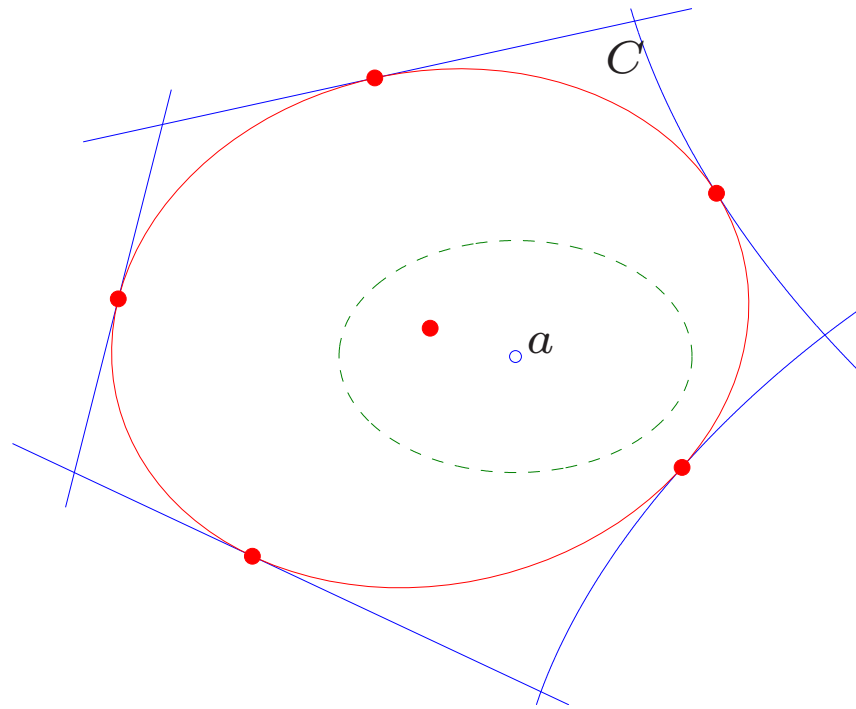
SDP formulation (variables $P \in \mathbf{S}^n$, $q \in \mathbf{R}^n$, $r, \tau_1, \dots, \tau_m \in \mathbf{R}$)

$$\text{maximize} \quad 1 - \text{tr}(SP) - 2a^T q - r$$

$$\text{subject to} \quad \begin{bmatrix} P & q \\ q^T & r - 1 \end{bmatrix} \succeq \tau_i \begin{bmatrix} A_i & b_i \\ b_i^T & c_i \end{bmatrix}, \quad \tau_i \geq 0 \quad i = 1, \dots, m$$
$$\begin{bmatrix} P & q \\ q^T & r \end{bmatrix} \succeq 0$$

optimal value is tight lower bound on $\text{prob}(X \in S)$

Example



- $a = \mathbf{E} X$; dashed line shows $\{x \mid (x - a)^T (S - aa^T)^{-1} (x - a) = 1\}$
- lower bound on $\mathbf{prob}(X \in C)$ is achieved by distribution shown in red
- ellipse is defined by $x^T P x + 2q^T x + r = 1$

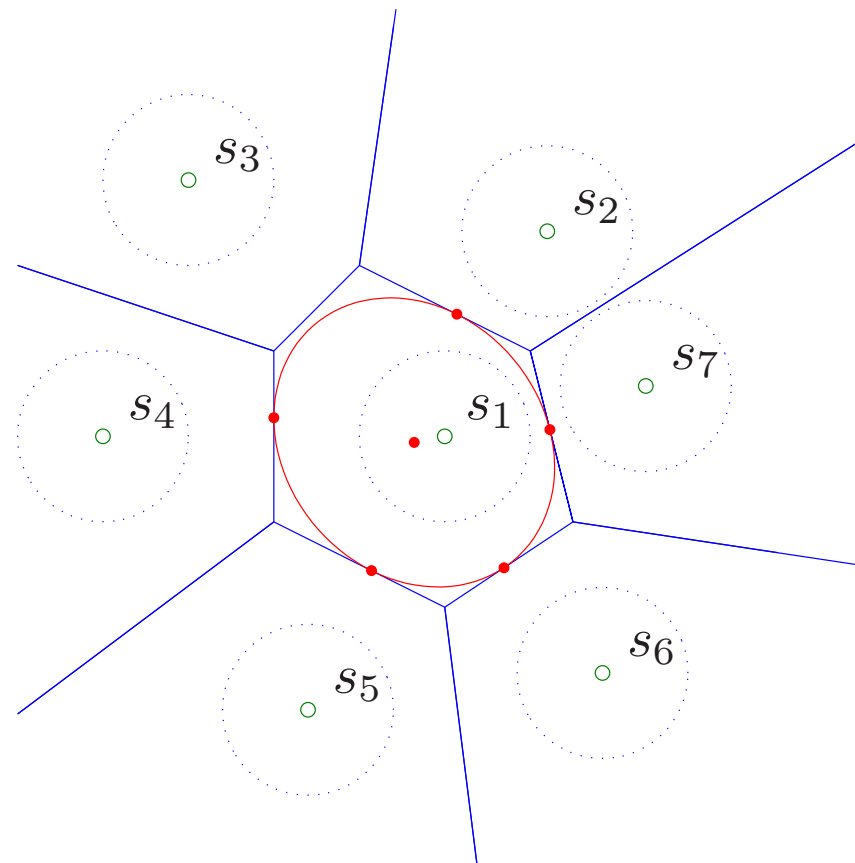
Detection example

$$x = s + v$$

- $x \in \mathbf{R}^n$: received signal
- s : transmitted signal $s \in \{s_1, s_2, \dots, s_N\}$ (one of N possible symbols)
- v : noise with $\mathbf{E} v = 0$, $\mathbf{E} v v^T = \sigma^2 I$

Detection problem: given observed value of x , estimate s

Example ($N = 7$): bound on probability of correct detection of s_1 is 0.205



dots: distribution with probability of correct detection 0.205

Cone programming duality

Primal and dual cone program

$$\begin{array}{ll} \text{P:} & \text{minimize} \quad c^T x \\ & \text{subject to} \quad Ax \preceq_K b \end{array}$$

$$\begin{array}{ll} \text{D:} & \text{maximize} \quad -b^T z \\ & \text{subject to} \quad A^T z + c = 0 \\ & \quad \quad \quad z \succeq_{K^*} 0 \end{array}$$

- optimal values are equal (if primal or dual is strictly feasible)
- dual inequality is with respect to the dual cone

$$K^* = \{z \mid x^T z \geq 0 \text{ for all } x \in K\}$$

- $K = K^*$ for linear, second-order cone, semidefinite programming

Applications: optimality conditions, sensitivity analysis, algorithms, . . .

Interior-point methods

- Newton's method
- barrier method
- primal-dual interior-point methods
- problem structure

Equality-constrained convex optimization

$$\begin{array}{ll} \text{minimize} & f(x) \\ \text{subject to} & Ax = b \end{array}$$

f twice continuously differentiable and convex

Optimality (Karush-Kuhn-Tucker or KKT) condition

$$\nabla f(x) + A^T y = 0, \quad Ax = b$$

Example: $f(x) = (1/2)x^T P x + q^T x + r$ with $P \succeq 0$

$$\begin{bmatrix} P & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -q \\ b \end{bmatrix}$$

a symmetric indefinite set of equations, known as a KKT system

Newton step

replace f with second-order approximation f_q at feasible $\hat{x}$:

$$\begin{array}{ll} \text{minimize} & f_q(x) \triangleq f(\hat{x}) + \nabla f(\hat{x})^T (x - \hat{x}) + \frac{1}{2}(x - \hat{x})^T \nabla^2 f(\hat{x})(x - \hat{x}) \\ \text{subject to} & Ax = b \end{array}$$

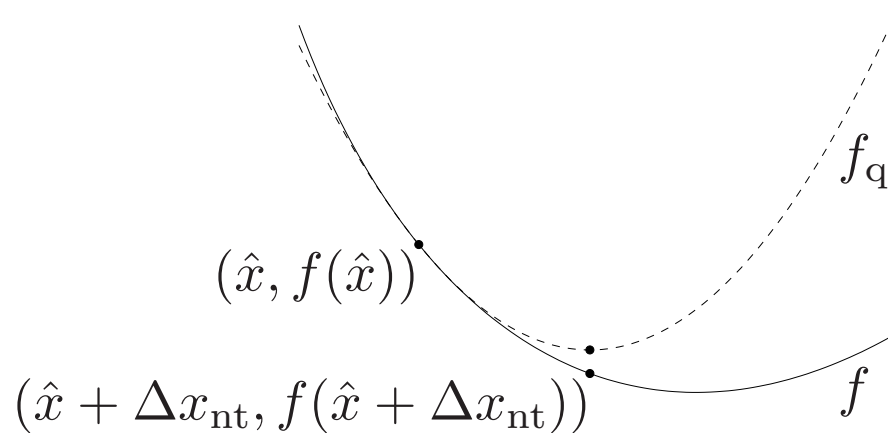
solution is $x = \hat{x} + \Delta x_{\text{nt}}$ with Δx_{nt} defined by

$$\begin{bmatrix} \nabla^2 f(\hat{x}) & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x_{\text{nt}} \\ w \end{bmatrix} = \begin{bmatrix} -\nabla f(\hat{x}) \\ 0 \end{bmatrix}$$

Δx_{nt} is called the **Newton step** at $\hat{x}$

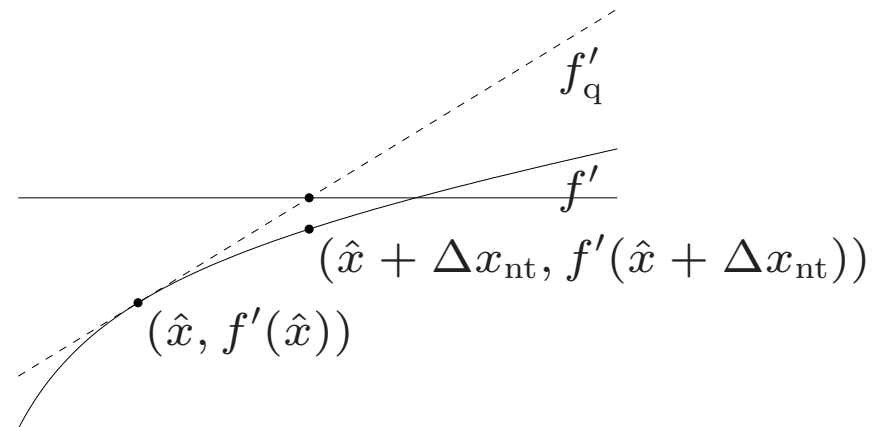
Interpretation (for unconstrained problem)

$\hat{x} + \Delta x_{\text{nt}}$ minimizes 2nd-order approximation f_q



$\hat{x} + \Delta x_{\text{nt}}$ solves linearized optimality condition

$$\begin{aligned} \nabla f_q(x) &= \nabla f(\hat{x}) + \nabla^2 f(\hat{x})(x - \hat{x}) \\ &= 0 \end{aligned}$$



Newton's algorithm

given starting point $x^{(0)} \in \text{dom } f$ with $Ax^{(0)} = b$, tolerance ϵ
repeat for $k = 0, 1, \dots$

1. compute Newton step Δx_{nt} at $x^{(k)}$ by solving

$$\begin{bmatrix} \nabla^2 f(x^{(k)}) & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x_{\text{nt}} \\ w \end{bmatrix} = \begin{bmatrix} -\nabla f(x^{(k)}) \\ 0 \end{bmatrix}$$

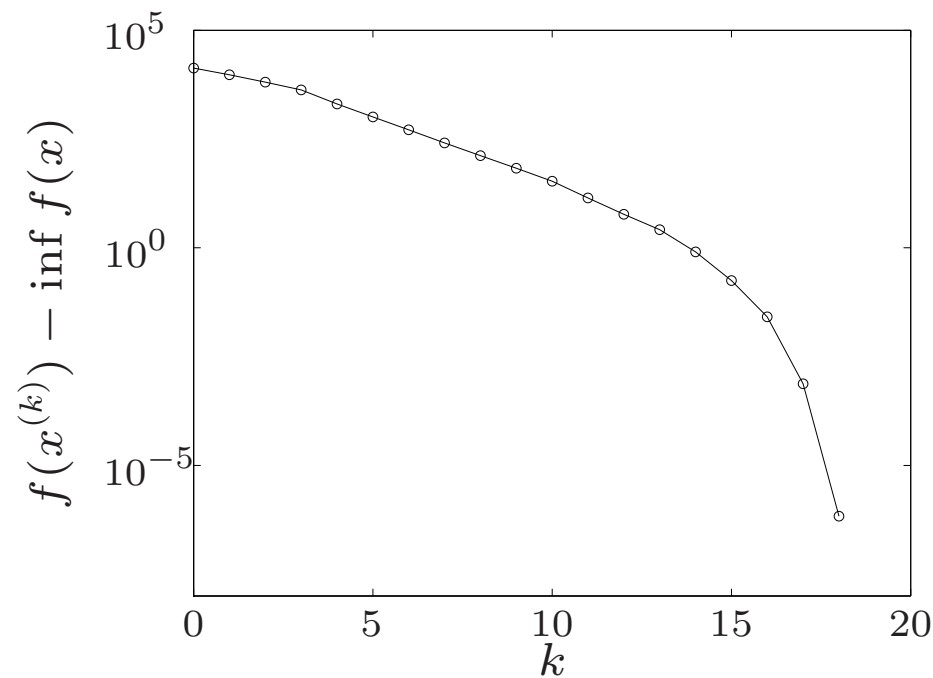
2. terminate if $-\nabla f(x^{(k)})^T \Delta x_{\text{nt}} \leq \epsilon$
3. $x^{(k+1)} = x^{(k)} + t \Delta x_{\text{nt}}$, with t determined by line search

Comments

- $\nabla f(x^{(k)})^T \Delta x_{\text{nt}}$ is directional derivative at $x^{(k)}$ in Newton direction
- line search needed to guarantee $f(x^{(k+1)}) < f(x^{(k)})$, global convergence

Example

$$f(x) = -\sum_{i=1}^n \log(1 - x_i^2) - \sum_{i=1}^m \log(b_i - a_i^T x) \quad (\text{with } n = 10^4, m = 10^5)$$



- high accuracy after small number of iterations
- fast asymptotic convergence

Classical convergence analysis

Assumptions (m, L are positive constants)

- f strongly convex: $\nabla^2 f(x) \succeq mI$
- $\nabla^2 f$ Lipschitz continuous: $\|\nabla^2 f(x) - \nabla^2 f(y)\|_2 \leq L\|x - y\|_2$

Summary: two regimes

- damped phase ($\|\nabla f(x)\|_2$ large): for some constant $\gamma > 0$

$$f(x^{(k+1)}) - f(x^{(k)}) \leq -\gamma$$

- quadratic convergence ($\|\nabla f(x)\|_2$ small)

$$\|\nabla f(x^{(k)})\|_2 \text{ decreases quadratically}$$

Self-concordant functions

Shortcomings of classical convergence analysis

- depends on unknown constants $(m, L, \dots)$
- bound is not affinely invariant, although Newton's method is

Analysis for self-concordant functions (Nesterov and Nemirovski, 1994)

- a convex function of one variable is self-concordant if

$$|f'''(x)| \leq 2f''(x)^{3/2} \quad \text{for all } x \in \text{dom } f$$

a function of several variables is s.c. if its restriction to lines is s.c.

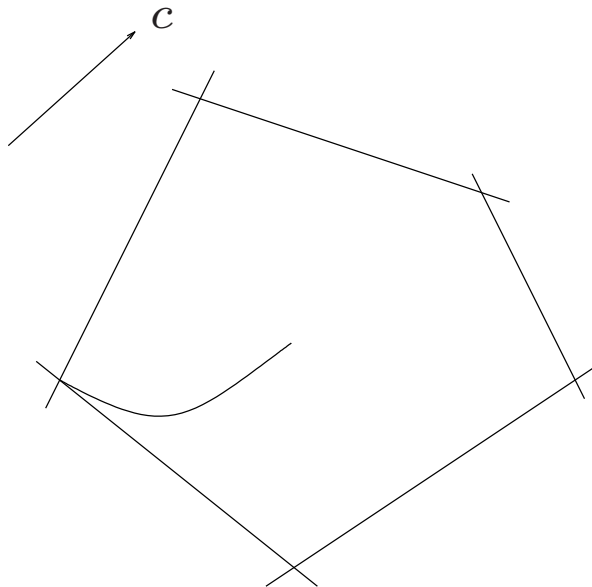
- analysis is affine-invariant, does not depend on unknown constants
- developed for complexity theory of interior-point methods

Interior-point methods

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & Ax = b\end{array}$$

functions f_i , $i = 0, 1, \dots, m$, are convex

Basic idea: follow 'central path' through interior feasible set to solution



General properties

- path-following mechanism relies on Newton's method
- every iteration requires solving a set of linear equations (KKT system)
- number of iterations small (10–50), fairly independent of problem size
- some versions known to have polynomial worst-case complexity

History

- introduced in 1950s and 1960s
- used in polynomial-time methods for linear programming (1980s)
- polynomial-time algorithms for general convex optimization (ca. 1990)

Reformulation via indicator function

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & Ax = b\end{array}$$

Reformulation

$$\begin{array}{ll}\text{minimize} & f_0(x) + \sum_{i=1}^m I_-(f_i(x)) \\ \text{subject to} & Ax = b\end{array}$$

where I_- is indicator function of $\mathbf{R}_-$:

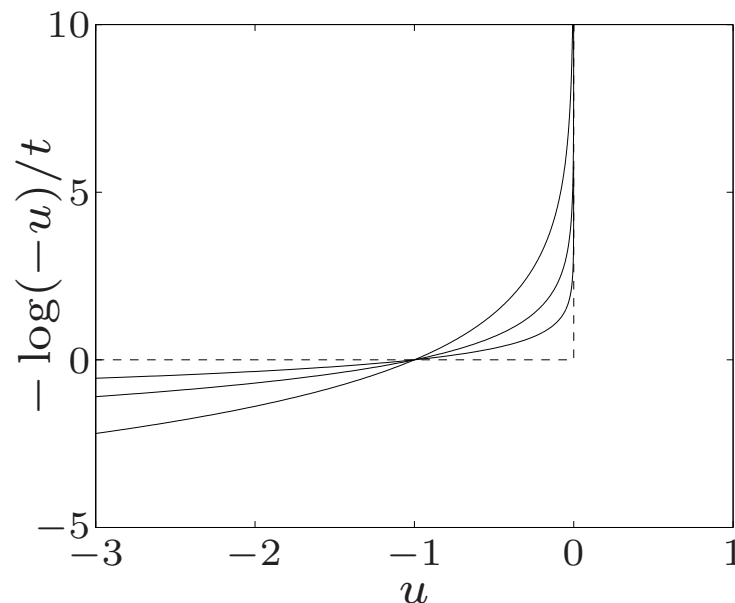
$$I_-(u) = 0 \quad \text{if } u \leq 0, \quad I_-(u) = \infty \quad \text{otherwise}$$

- reformulated problem has no inequality constraints
- however, objective function is not differentiable

Approximation via logarithmic barrier

$$\begin{array}{ll}\text{minimize} & f_0(x) - \frac{1}{t} \sum_{i=1}^m \log(-f_i(x)) \\ \text{subject to} & Ax = b\end{array}$$

- for $t > 0$, $-(1/t) \log(-u)$ is a smooth approximation of I_-
- approximation improves as $t \rightarrow \infty$



Logarithmic barrier function

$$\phi(x) = - \sum_{i=1}^m \log(-f_i(x))$$

with $\text{dom } \phi = \{x \mid f_1(x) < 0, \dots, f_m(x) < 0\}$

- convex (follows from composition rules and convexity of f_i)
- twice continuously differentiable, with derivatives

$$\nabla \phi(x) = \sum_{i=1}^m \frac{1}{-f_i(x)} \nabla f_i(x)$$

$$\nabla^2 \phi(x) = \sum_{i=1}^m \frac{1}{f_i(x)^2} \nabla f_i(x) \nabla f_i(x)^T + \sum_{i=1}^m \frac{1}{-f_i(x)} \nabla^2 f_i(x)$$

Central path

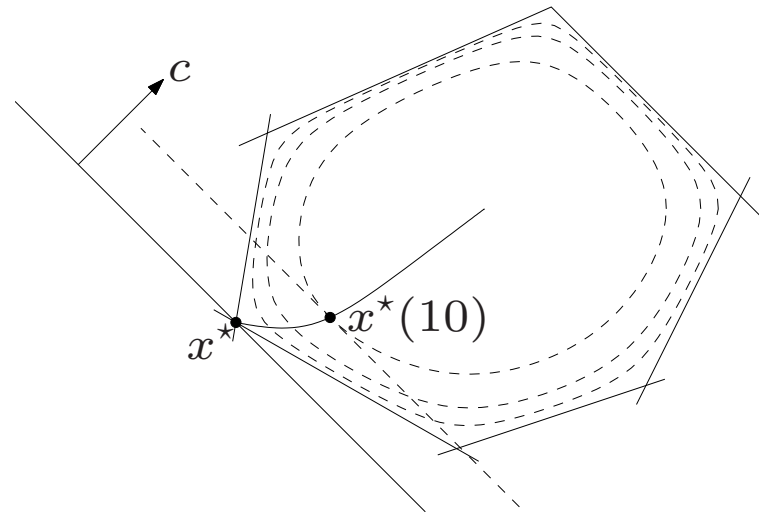
central path is $\{x^*(t) \mid t > 0\}$, where $x^*(t)$ is the solution of

$$\begin{array}{ll} \text{minimize} & t f_0(x) + \phi(x) \\ \text{subject to} & Ax = b \end{array}$$

Example: central path for an LP

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & a_i^T x \leq b_i, \quad i = 1, \dots, 6 \end{array}$$

hyperplane $c^T x = c^T x^*(t)$ is tangent to level curve of ϕ through $x^*(t)$



Barrier method

given strictly feasible x , $t := t^{(0)} > 0$, $\mu > 1$, tolerance $\epsilon > 0$
repeat:

1. *Centering step.* Compute $x^*(t)$ and set $x := x^*(t)$
2. *Stopping criterion.* Terminate if $m/t < \epsilon$
3. *Increase t .* $t := \mu t$

- stopping criterion $m/t \leq \epsilon$ guarantees

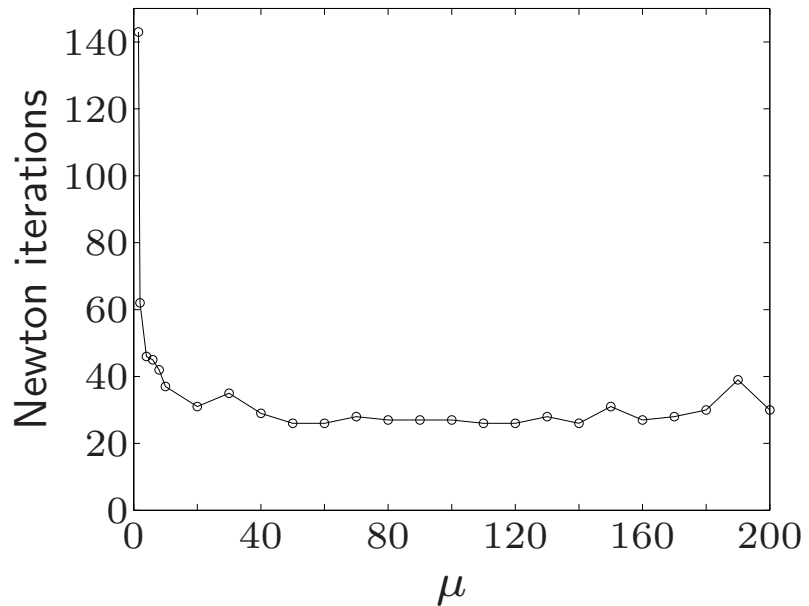
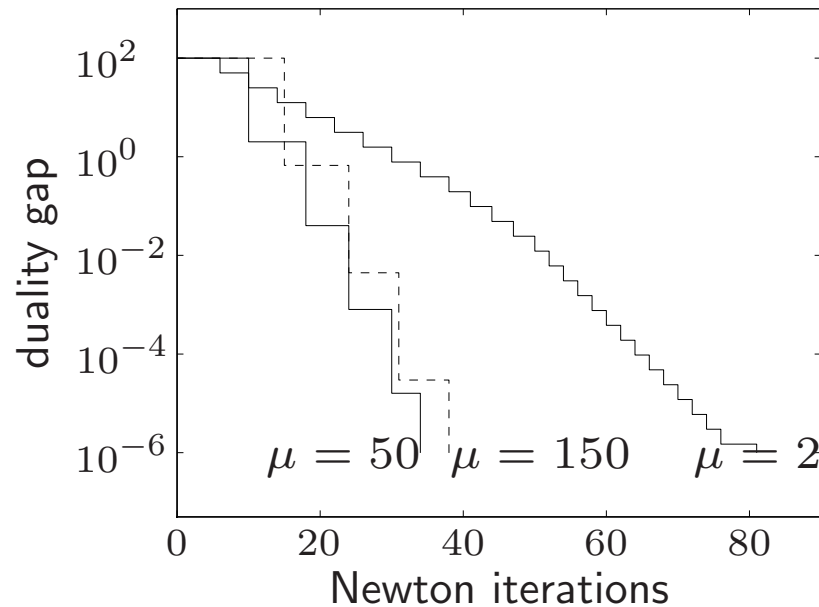
$$f_0(x) - \text{optimal value} \leq \epsilon$$

(follows from duality)

- typical value of μ is 10–20
- several heuristics for choice of $t^{(0)}$
- centering usually done using Newton's method, starting at current x

Example: Inequality form LP

$m = 100$ inequalities, $n = 50$ variables

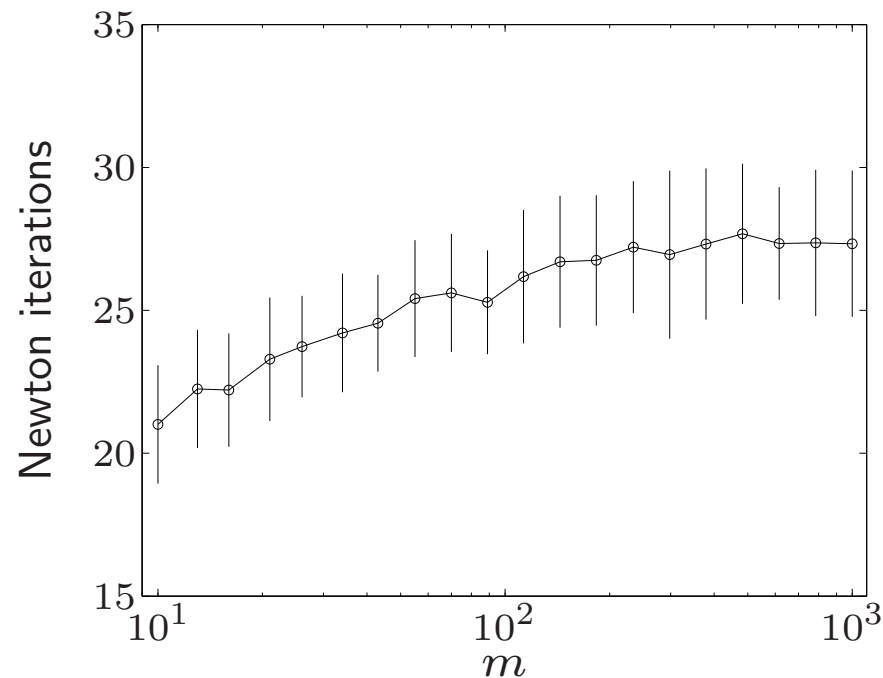


- starts with x on central path ($t^{(0)} = 1$, duality gap 100)
- terminates when $t = 10^8$ (gap $m/t = 10^{-6}$)
- total number of Newton iterations not very sensitive for $\mu \geq 10$

Family of standard LPs

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & Ax = b, \quad x \succeq 0\end{array}$$

$A \in \mathbf{R}^{m \times 2m}$; for each m , solve 100 randomly generated instances



number of iterations grows very slowly as m ranges over a 100 : 1 ratio

Second-order cone programming

$$\begin{array}{ll}\text{minimize} & f^T x \\ \text{subject to} & \|A_i x + b_i\|_2 \leq c_i^T x + d_i, \quad i = 1, \dots, m\end{array}$$

Logarithmic barrier function

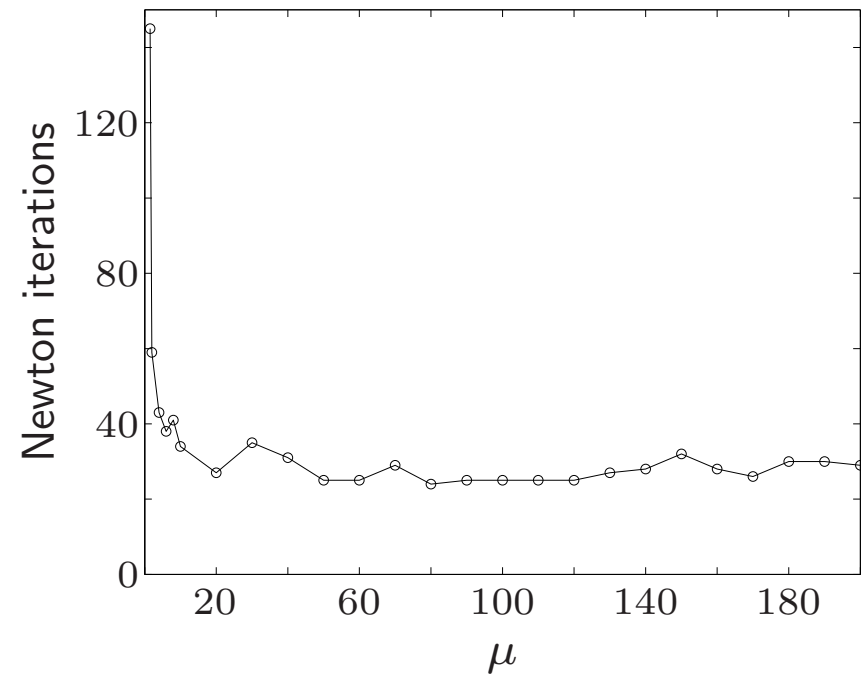
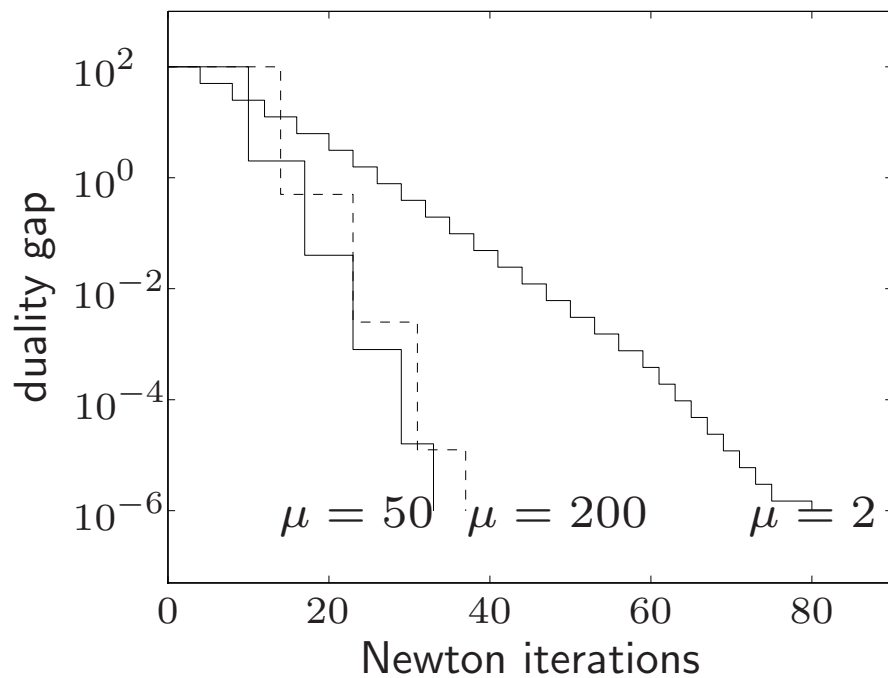
$$\phi(x) = - \sum_{i=1}^m \log \left((c_i^T x + d_i)^2 - \|A_i x + b_i\|_2^2 \right)$$

- a convex function
- $\log(v^2 - u^T u)$ is ‘logarithm’ for 2nd-order cone $\{(u, v) \mid \|u\|_2 \leq v\}$

Barrier method: follows central path $x^*(t) = \operatorname{argmin}(t f^T x + \phi(x))$

Example

50 variables, 50 second-order cone constraints in $\mathbf{R}^6$



Semidefinite programming

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & x_1 A_1 + \cdots + x_n A_n \preceq B\end{array}$$

Logarithmic barrier function

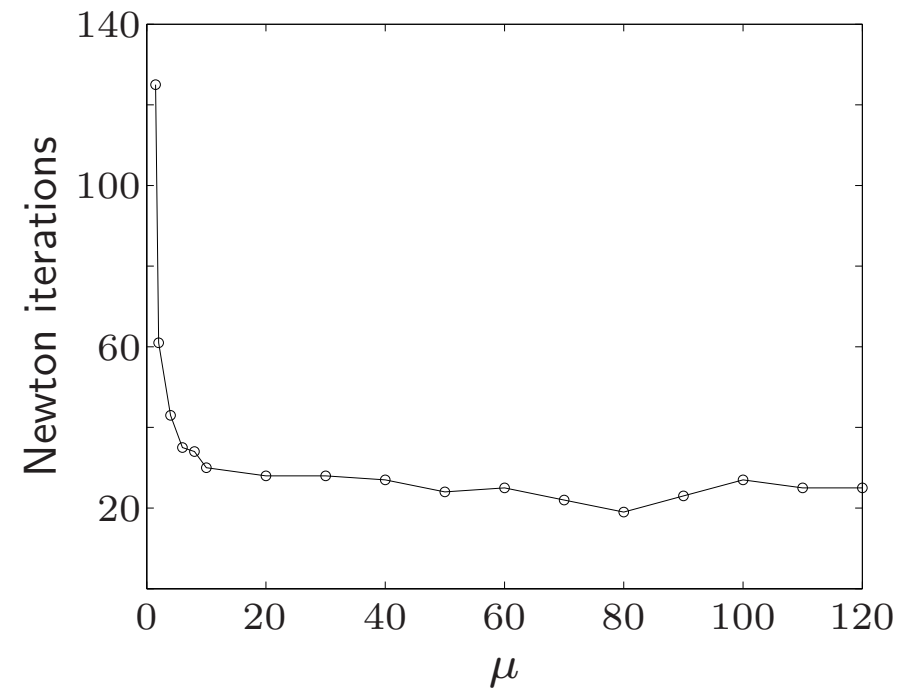
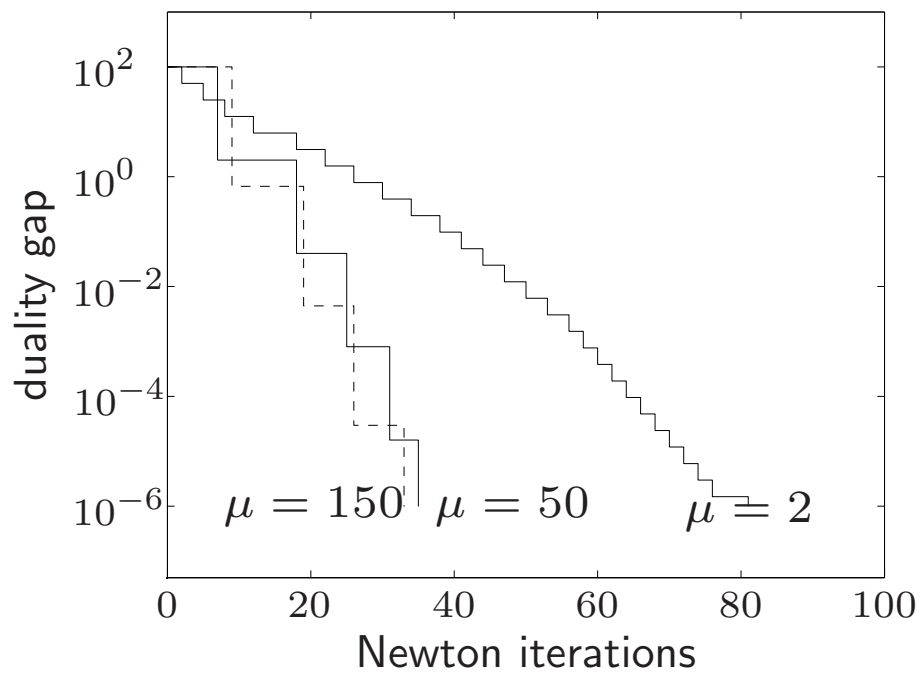
$$\phi(x) = -\log \det(B - x_1 A_1 - \cdots - x_n A_n)$$

- a convex function
- $\log \det X$ is 'logarithm' for p.s.d. cone

Barrier method: follows central path $x^*(t) = \operatorname{argmin}(tf^T x + \phi(x))$

Example

100 variables, one linear matrix inequality in $\mathbf{S}^{100}$



Complexity of barrier method

Iteration complexity

- can be bounded by polynomial function of problem dimensions (with correct formulation, barrier function)
- examples: $O(\sqrt{m})$ iteration bound for LP or SOCP with m inequalities, SDP with constraint of order m
- proofs rely on theory of Newton's method for self-concordant functions
- in practice: #iterations roughly constant as a function of problem size

Linear algebra complexity

dominated by solution of Newton system

Primal-dual interior-point methods

Similarities with barrier method

- follow the same central path
- linear algebra (KKT system) per iteration is similar

Differences

- faster and more robust
- update primal and dual variables in each step
- no distinction between inner (centering) and outer iterations
- include heuristics for adaptive choice of barrier parameter t
- can start at infeasible points
- often exhibit superlinear asymptotic convergence

Software implementations

General-purpose software for nonlinear convex optimization

- several high-quality packages (MOSEK, Sedumi, SDPT3, . . .)
- exploit sparsity to achieve scalability

Customized implementations

- can exploit non-sparse types of problem structure
- often orders of magnitude faster than general-purpose solvers

Example: ℓ_1 -regularized least-squares

$$\text{minimize} \quad \|Ax - b\|_2^2 + \|x\|_1$$

A is $m \times n$ (with $m \leq n$) and dense

Quadratic program formulation

$$\begin{aligned} &\text{minimize} \quad \|Ax - b\|_2^2 + \mathbf{1}^T u \\ &\text{subject to} \quad -u \preceq x \preceq u \end{aligned}$$

- coefficient of Newton system in interior-point method is

$$\begin{bmatrix} A^T A & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} D_1 + D_2 & D_2 - D_1 \\ D_2 - D_1 & D_1 + D_2 \end{bmatrix} \quad (D_1, D_2 \text{ positive diagonal})$$

- very expensive ($O(n^3)$) for large n

Customized implementation

- can reduce Newton equation to solution of a system

$$(AD^{-1}A^T + I)\Delta u = r$$

- cost per iteration is $O(m^2n)$

Comparison (seconds on 3.2Ghz machine)

m	n	custom	general-purpose
50	100	0.02	0.05
50	200	0.03	0.17
100	1000	0.32	10.6
100	2000	0.71	76.9
500	1000	2.5	11.2
500	2000	5.5	79.8

general-purpose solver is MOSEK

First-order methods

- gradient method
- Nesterov's gradient methods
- extensions

Gradient method

to minimize a convex differentiable function f : choose $x^{(0)}$ and repeat

$$x^{(k)} = x^{(k-1)} - t_k \nabla f(x^{(k-1)}), \quad k = 1, 2, \dots$$

t_k is step size (fixed or determined by backtracking line search)

Classical convergence result

- assume ∇f Lipschitz continuous ($\|\nabla f(x) - \nabla f(y)\|_2 \leq L\|x - y\|_2$)
- error decreases as $1/k$, hence

$$O\left(\frac{1}{\epsilon}\right) \text{ iterations}$$

needed to reach accuracy $f(x^{(k)}) - f^* \leq \epsilon$

Nesterov's gradient method

choose $x^{(0)}$; take $x^{(1)} = x^{(0)} - t_1 \nabla f(x^{(0)})$ and for $k \geq 2$

$$y^{(k)} = x^{(k-1)} + \frac{k-2}{k+1}(x^{(k-1)} - x^{(k-2)})$$

$$x^{(k)} = y^{(k)} - t_k \nabla f(y^{(k)})$$

- gradient method with 'extrapolation'
- if f has Lipschitz continuous gradient, error decreases as $1/k^2$; hence

$$O\left(\frac{1}{\sqrt{\epsilon}}\right) \text{ iterations}$$

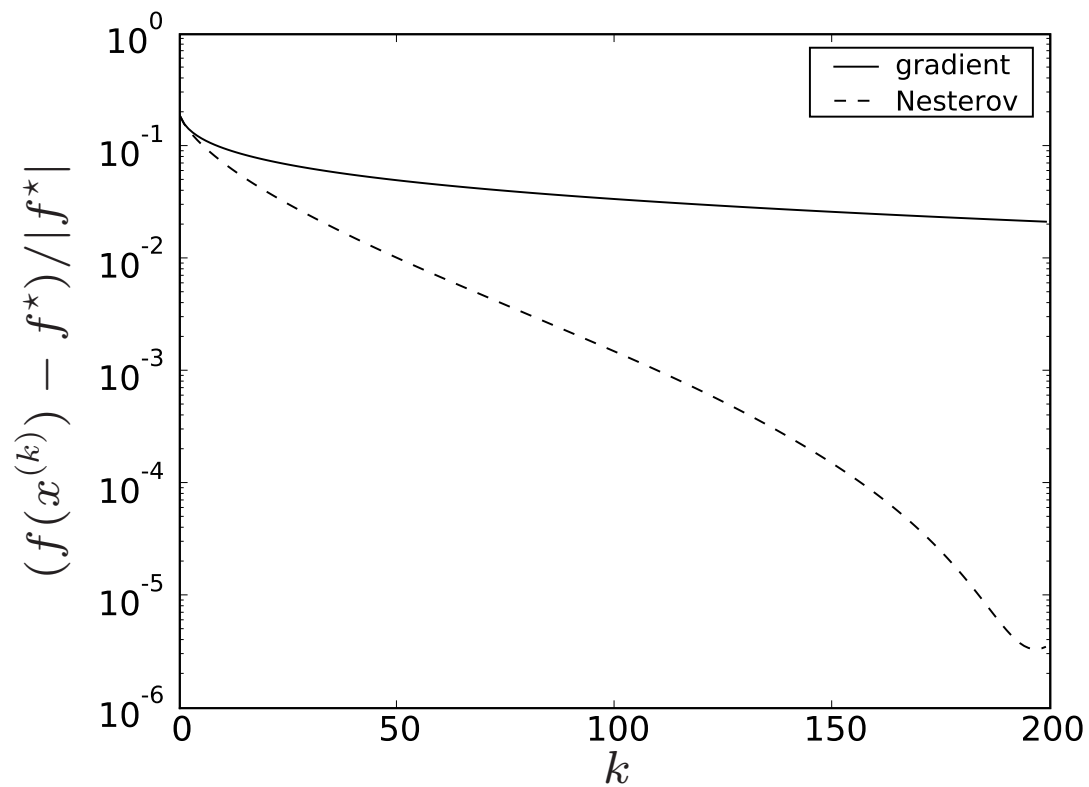
needed to reach accuracy $f(x^{(k)}) - f^* \leq \epsilon$

- many variations; first one published in 1983

Example

$$\text{minimize} \quad \log \sum_{i=1}^m \exp(a_i^T x + b_i)$$

randomly generated data with $m = 2000$, $n = 1000$, fixed step size



Interpretation of gradient update

$$\begin{aligned}x^{(k)} &= x^{(k-1)} - t_k \nabla f(x^{(k-1)}) \\&= \operatorname{argmin}_z \left(\nabla f(x^{(k-1)})^T z + \frac{1}{t_k} \|z - x^{(k-1)}\|_2^2 \right)\end{aligned}$$

Interpretation

$x^{(k)}$ minimizes

$$f(x^{(k-1)}) + \nabla f(x^{(k-1)})^T (z - x^{(k-1)}) + \frac{1}{t_k} \|z - x^{(k-1)}\|_2^2$$

a simple quadratic model of f at $x^{(k-1)}$

Projected gradient method

$$\begin{array}{ll} \text{minimize} & f(x) \\ \text{subject to} & x \in C \end{array}$$

f convex, C a closed convex set

$$\begin{aligned} x^{(k)} &= \operatorname{argmin}_{z \in C} \left(\nabla f(x^{(k-1)})^T z + \frac{1}{t_k} \|z - x^{(k-1)}\|_2^2 \right) \\ &= P_C \left(x^{(k-1)} - t_k \nabla f(x^{(k-1)}) \right) \end{aligned}$$

- useful if projection P_C on C is inexpensive (*e.g.*, box constraints)
- similar convergence result as for basic gradient algorithm
- can be used in fast Nesterov-type gradient methods

Nonsmooth components

$$\text{minimize } f(x) + g(x)$$

f, g convex, with f differentiable, g nondifferentiable

$$\begin{aligned} x^{(k)} &= \operatorname{argmin}_z \left(\nabla f(x^{(k-1)})^T z + g(x) + \frac{1}{t_k} \|z - x^{(k-1)}\|_2^2 \right) \\ &= \operatorname{argmin}_z \left(\frac{1}{2t_k} \left\| z - x^{(k-1)} + t_k \nabla f(x^{(k-1)}) \right\|_2^2 + g(z) \right) \\ &\triangleq S_{t_k} \left(x^{(k-1)} - t_k \nabla f(x^{(k-1)}) \right) \end{aligned}$$

- gradient step for f followed by ‘thresholding’ operation S_t
- useful if thresholding is inexpensive (*e.g.*, because g is separable)
- similar convergence result as basic gradient method

Example: ℓ_1 -norm regularization

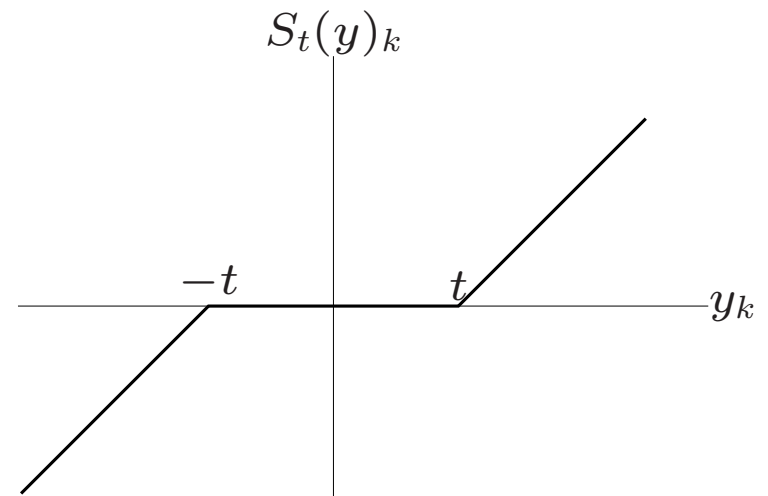
$$\text{minimize } f(x) + \|x\|_1$$

f convex and differentiable

Thresholding operator

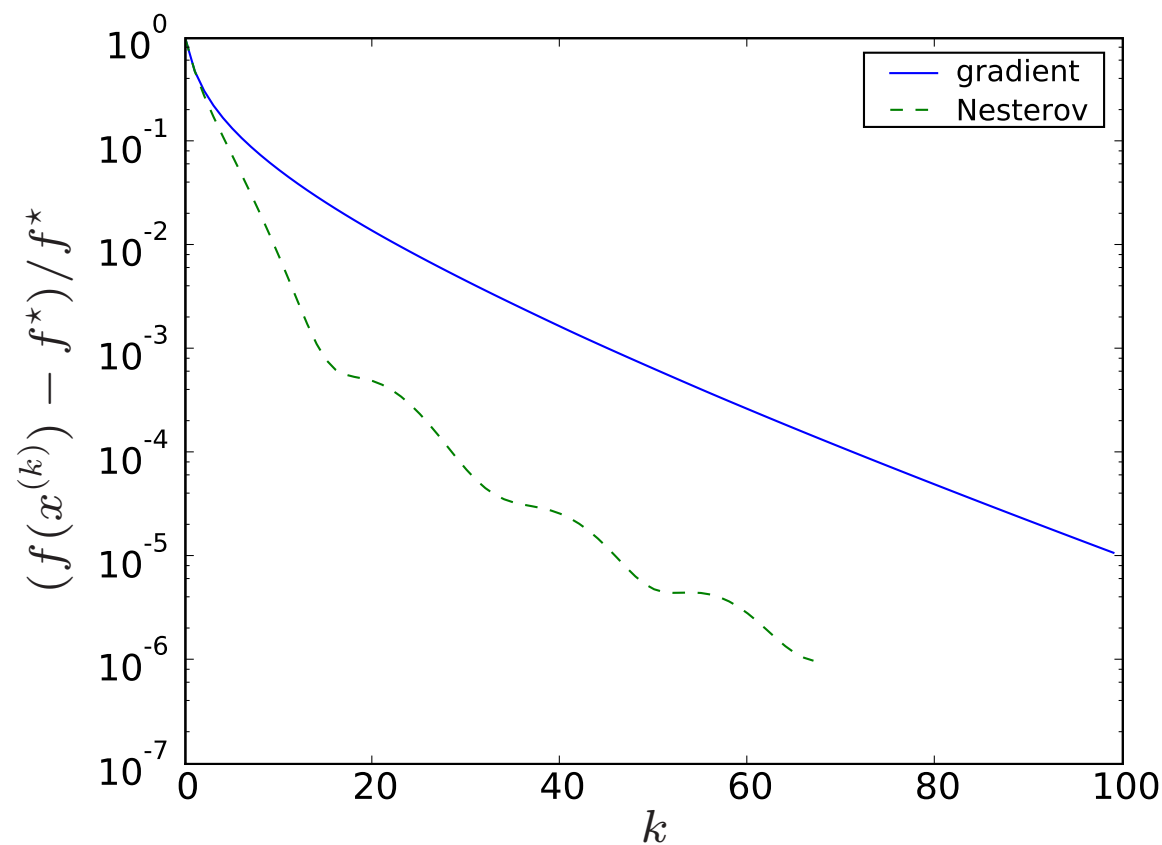
$$S_t(y) = \operatorname{argmin}_z \left(\frac{1}{2t} \|z - y\|_2^2 + \|z\|_1 \right)$$

$$S_t(y)_k = \begin{cases} y_k - t & y_k \geq t \\ 0 & -t \leq y_k \leq t \\ y_k + t & y_k \leq -t \end{cases}$$



ℓ_1 -Norm regularized least-squares

$$\text{minimize } \frac{1}{2}\|Ax - b\|_2^2 + \|x\|_1$$



randomly generated $A \in \mathbf{R}^{2000 \times 1000}$; fixed step

Summary: Advances in convex optimization

Theory

new problem classes, robust optimization, convex relaxations, . . .

Applications

new applications in different fields; surprisingly many discovered recently

Algorithms and software

- high-quality general-purpose implementations of interior-point methods
- software packages for convex modeling
- new first-order methods